Machine Programming

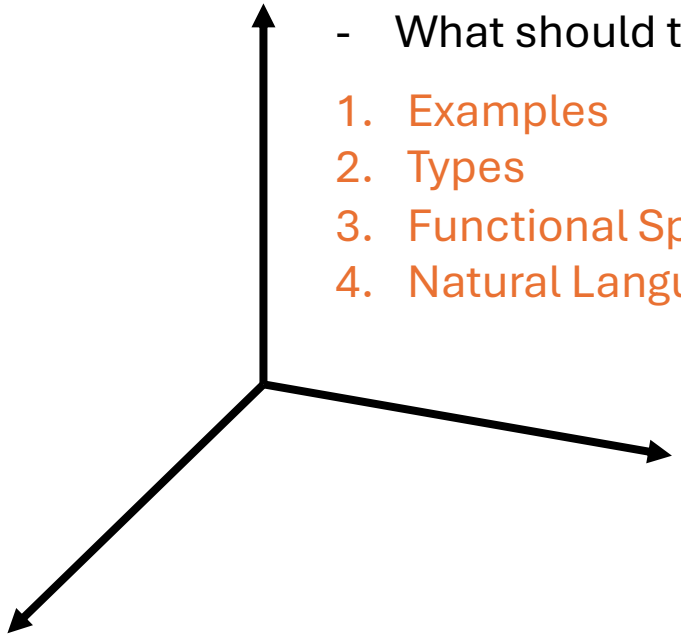
Lecture 15 – PL Kaleidoscope I: Formal Verification & Theorem Provers

Ziyang Li

The Course So Far

Behavioral Specification
- What should the program do?

1. Examples
2. Types
3. Functional Specifications
4. Natural Language



Synthesis Strategy
- How do we find such a program?

Language Models
- Pre-training, Post-training, etc.
- Prompting
- Iterative refinement
- Agentic frameworks

Structural Specification
- What is the space of the programs?

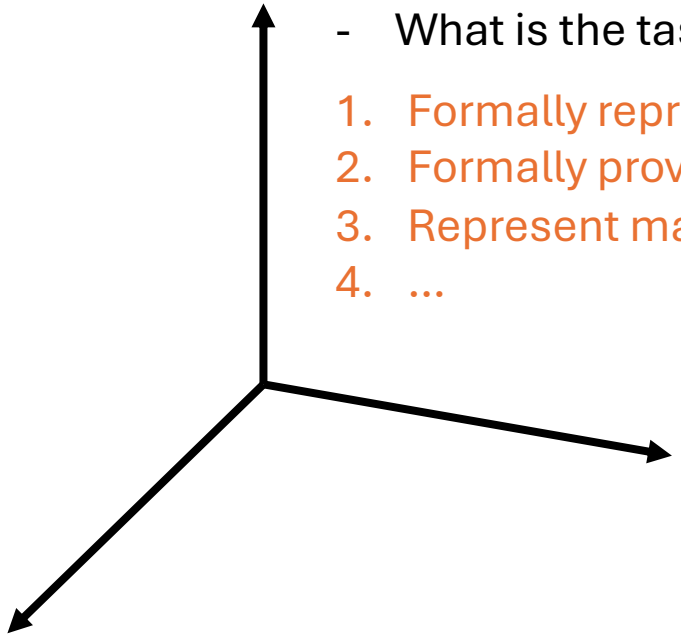
General Purpose Programming Language
Python / Java / C / Rust / …

Domain Specific Languages

The discussion today

Behavioral Specification
- What should the program do?
- What is the task for producing this program?

1. Formally represent mathematics: **Autoformalization**
2. Formally prove a theorem: **Proof Search**
3. Represent math in different formalisms: **Proof Porting**
4. …



Synthesis Strategy
- How do we find such a program?

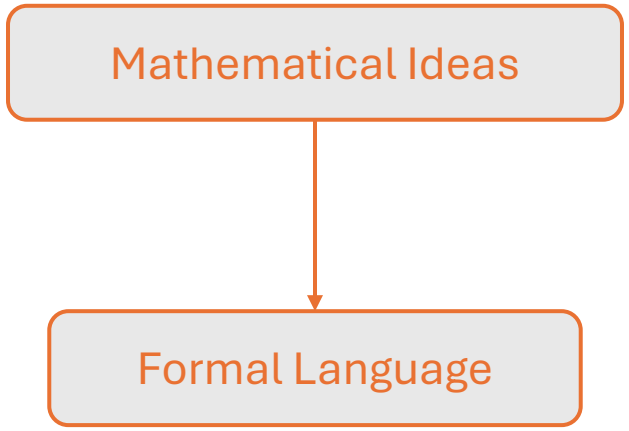
Language Models
- Pre-training, Post-training, etc.
- Prompting
- Iterative refinement
- Agentic frameworks

Structural Specification
- What is the space of the programs?

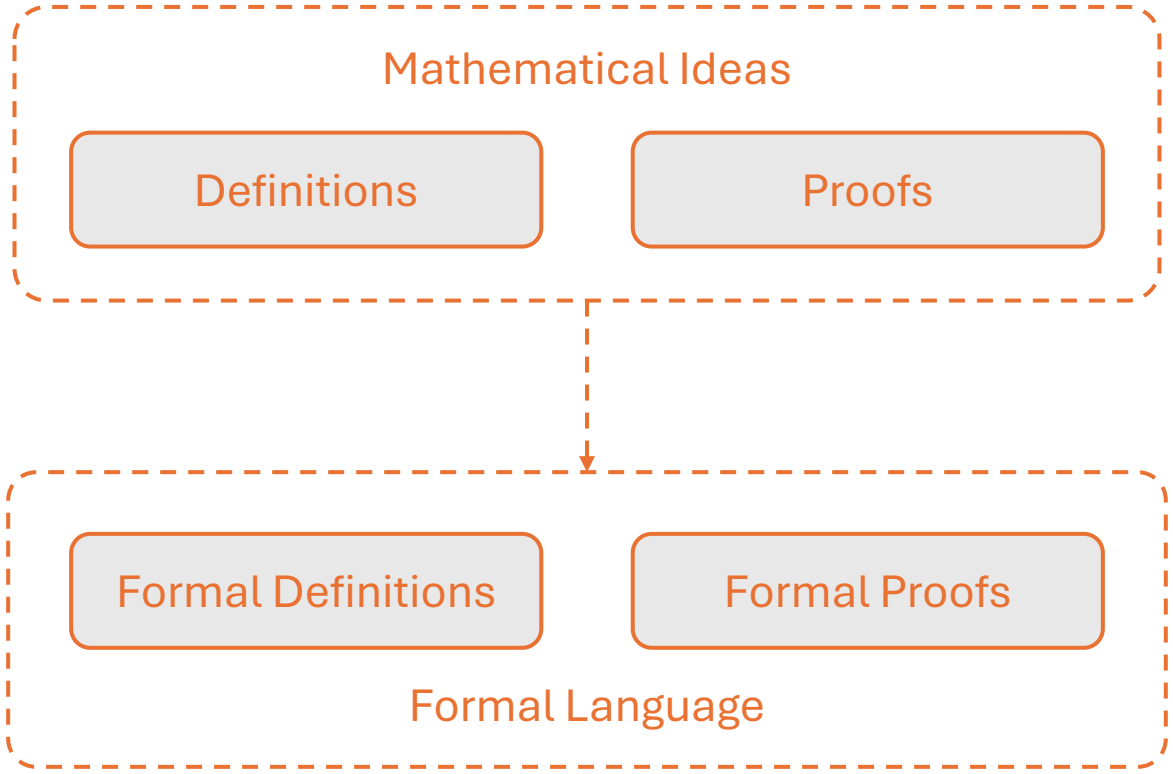
General Purpose Programming Language
Python / Java / C / Rust / …

Domain Specific Languages
Lean/Rocq/Isabelle/Z3 for theorem proving

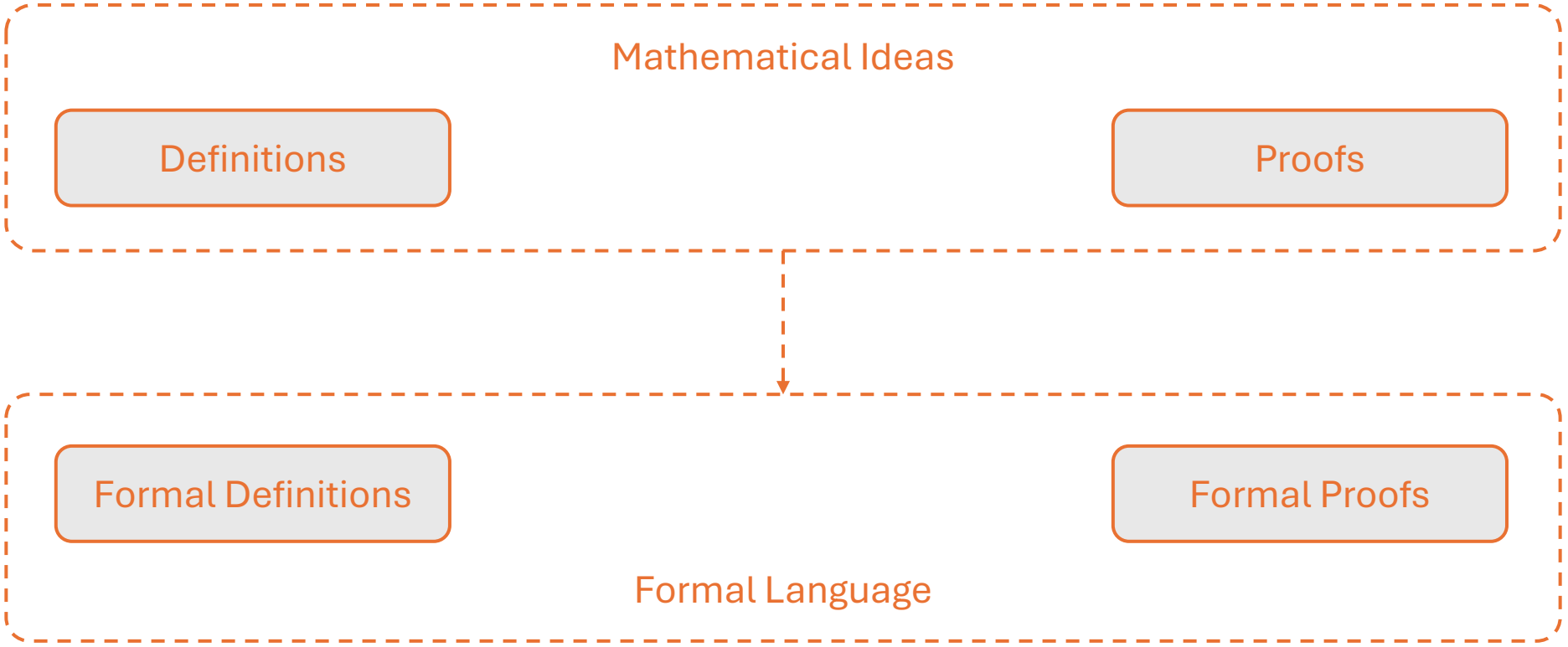
Formal Verification for Mathematics



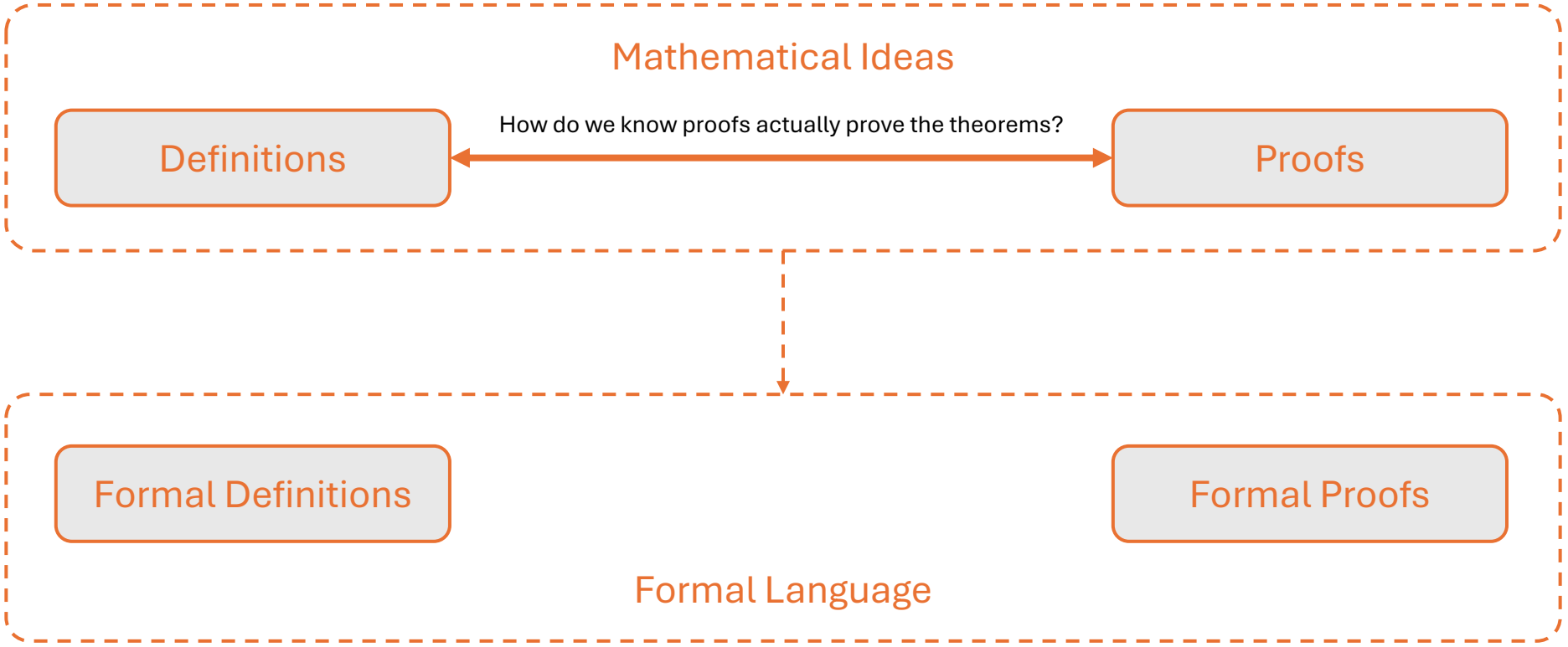
Central Thesis



Central Thesis



Central Thesis



A Solution of the P versus NP Problem

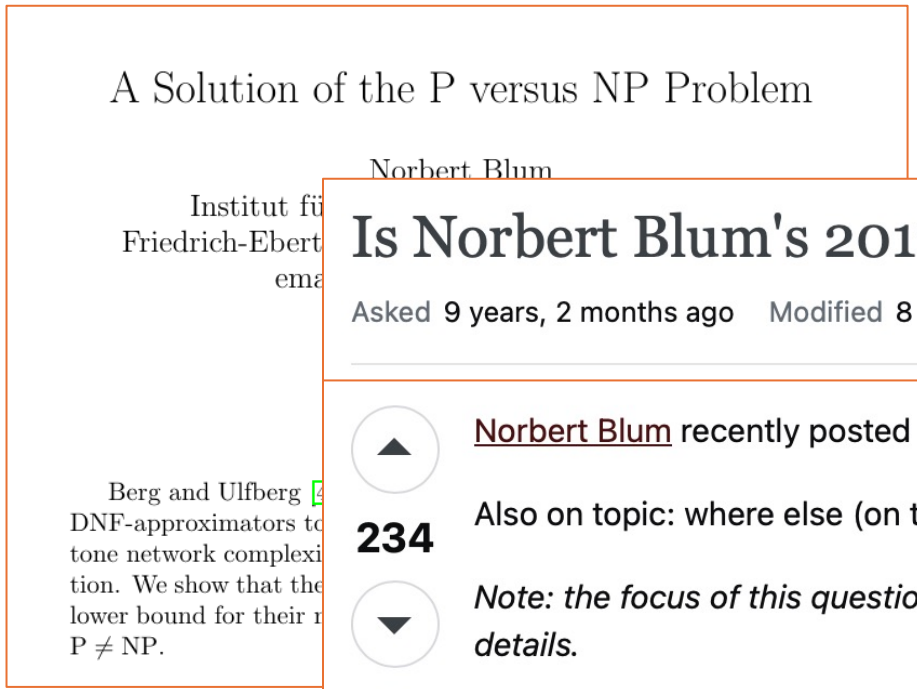
Norbert Blum
Institut für Informatik, Universität Bonn
Friedrich-Ebert-Allee 144, D-53113 Bonn, Germany
email: blum@cs.uni-bonn.de

August 14, 2017

Abstract

Berg and Ulfberg [4] and Amano and Maruoka [2] have used CNF-DNF-approximators to prove exponential lower bounds for the monotone network complexity of the clique function and of Andreev's function. We show that these approximators can be used to prove the same lower bound for their non-monotone network complexity. This implies $P \neq NP$.

https://arxiv.org/abs/1708.03486v1



A Solution of the P versus NP Problem

Norbert Blum

Institut fü
Friedrich-Ebert
ema

Berg and Ulfberg
DNF-approximators to
tone network complexi
tion. We show that the
lower bound for their
P ≠ NP.

Is Norbert Blum's 2017 proof that $P \neq NP$ correct?

Asked 9 years, 2 months ago Modified 8 years, 1 month ago Viewed 123k times

234

Norbert Blum recently posted a 38-page proof that $P \neq NP$. Is it correct?

Also on topic: where else (on the internet) is its correctness being discussed?

Note: the focus of this question text has changed over time. See question comments for details.

cc.complexity-theory np-hardness complexity-classes

https://cstheory.stackexchange.com/questions/38803

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DNF
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11 Answers

Sorted by: Highest score (default)

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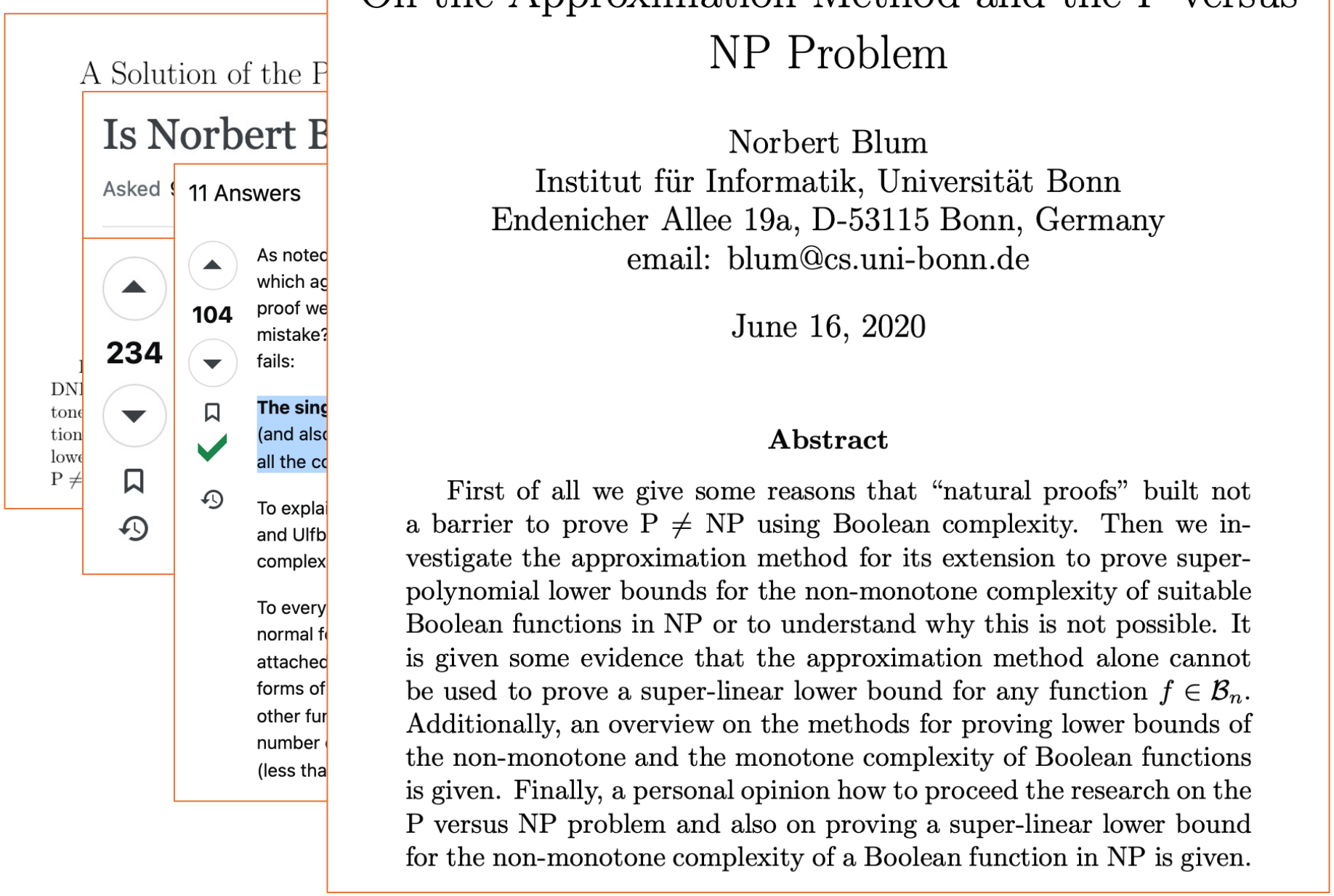
As noted here before, Tardos' example clearly refutes the proof; it gives a monotone function, which agrees with CLIQUE on T0 and T1, but which lies in P. This would not be possible if the proof were correct, since the proof applies to this case too. However, can we pinpoint the mistake? Here is, from a post on the lipton's blog, what seems to be the place where the proof fails:

The single error is one subtle point in the proof of Theorem 6, namely in Step 1, on page 31 (and also 33, where the dual case is discussed) - a seemingly obvious claim that C'_g contains all the corresponding clauses contained in $CNF'(g)$ etc, seems wrong.

To explain this in more detail, we need to go into the proof and approximation method of Berg and Ulfberg, which restates the Razborov's original proof of the exponential monotone complexity for CLIQUE in terms of DNF/CNF switches. This is how I see it:

To every node/gate g of a logic circuit β (containing binary OR/AND gates only), a conjunctive normal form $CNF(g)$, a disjunctive normal form $DNF(g)$, and approximators C^k_g and D^r_g are attached. CNF and DNF are simply the corresponding disjunctive and conjunctive normal forms of the gate output. D^r_g and C^k_g are also disjunctive and conjunctive forms, but of some other functions, "approximating" the gate output. They are however required to have bounded number of variables in each monomial for D^r_g (less than a constant r) and in each clause for C^k_g (less than a constant k).

https://cstheory.stackexchange.com/questions/38803



On the Approximation Method and the P versus NP Problem

Norbert Blum
Institut für Informatik, Universität Bonn
Endenicher Allee 19a, D-53115 Bonn, Germany
email: blum@cs.uni-bonn.de

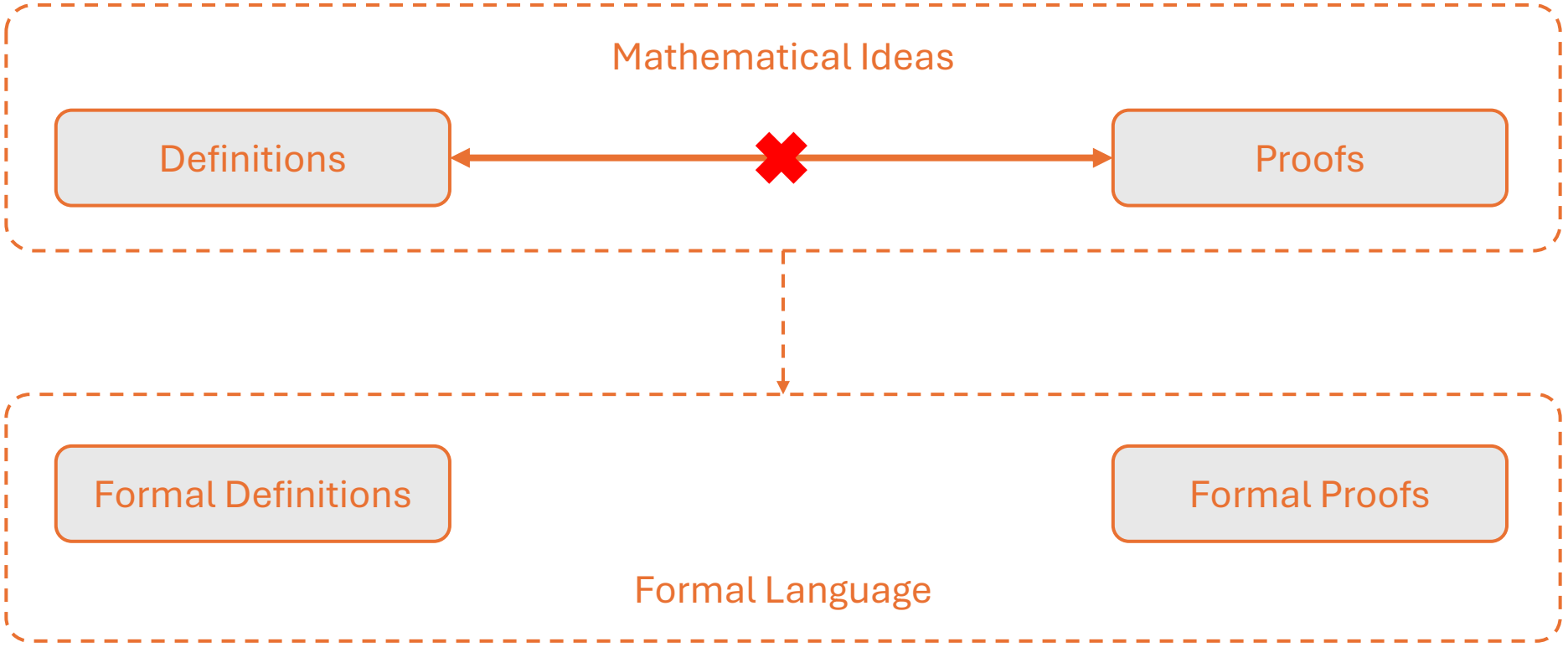
June 16, 2020

Abstract

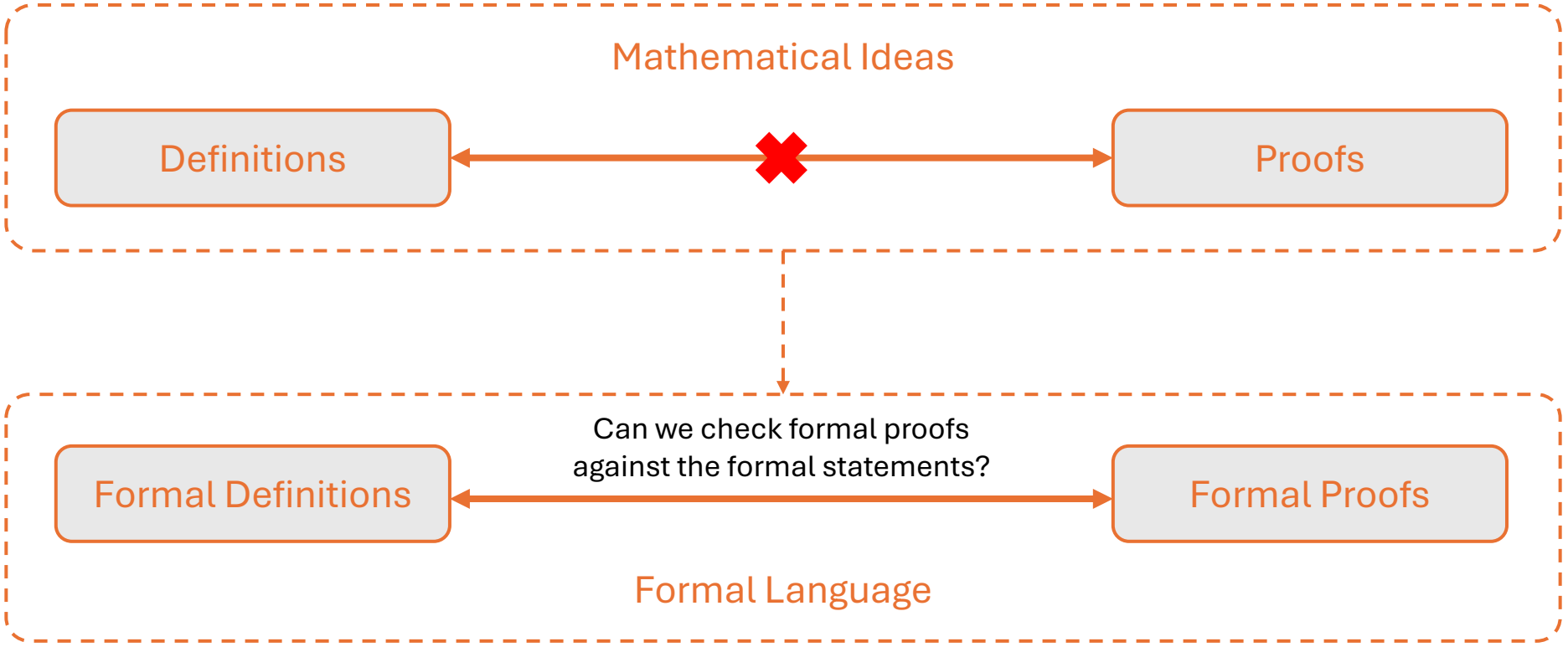
First of all we give some reasons that "natural proofs" built not a barrier to prove P $\neq$ NP using Boolean complexity. Then we investigate the approximation method for its extension to prove super-polynomial lower bounds for the non-monotone complexity of suitable Boolean functions in NP or to understand why this is not possible. It is given some evidence that the approximation method alone cannot be used to prove a super-linear lower bound for any function $f \in \mathcal{B}_n$. Additionally, an overview on the methods for proving lower bounds of the non-monotone and the monotone complexity of Boolean functions is given. Finally, a personal opinion how to proceed the research on the P versus NP problem and also on proving a super-linear lower bound for the non-monotone complexity of a Boolean function in NP is given.

https://arxiv.org/abs/1708.03486

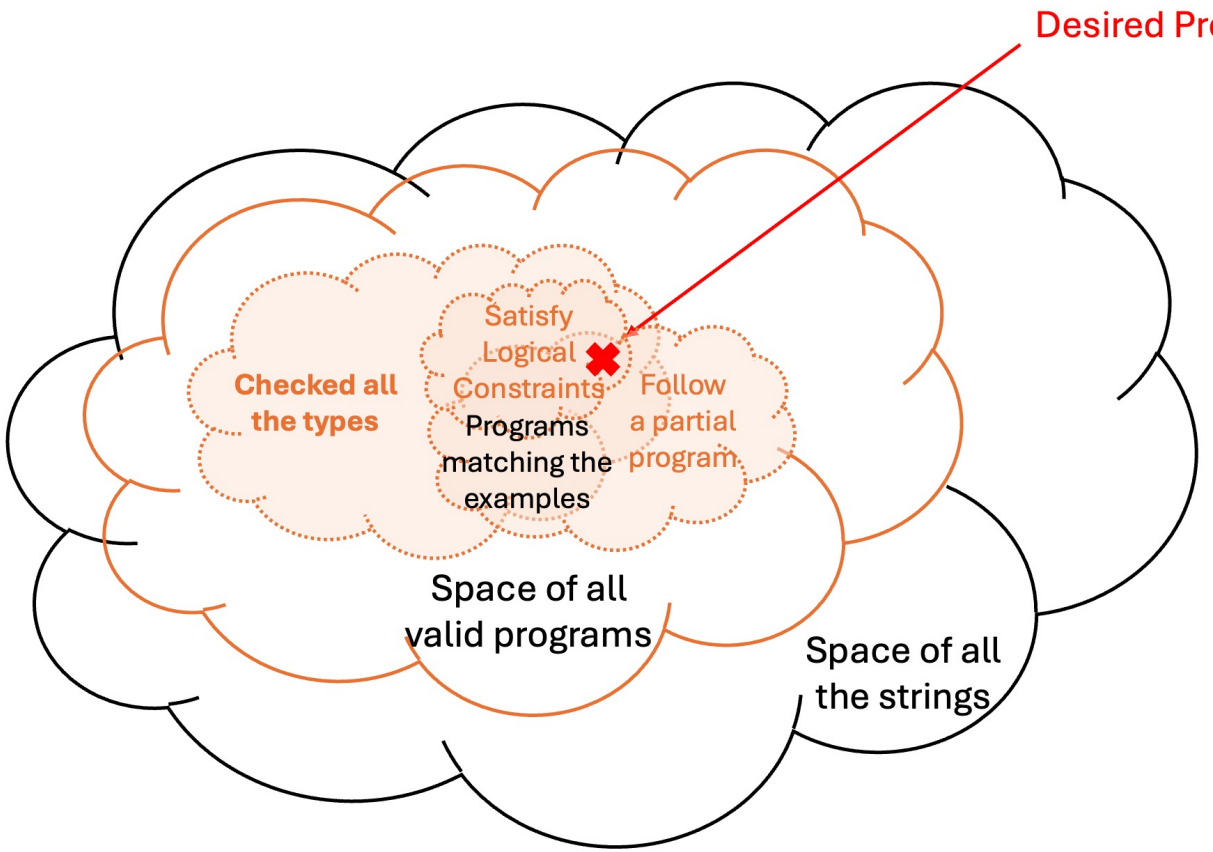
Central Thesis



Central Thesis



High Level Picture



Type Checking

Type checking a Java program...

```java
class Area {
  static int area(int w, int h) {
    return w * h;
  }
}
```

```
$ javac Area.java
✓ no errors
```

```java
class Area {
  static int area(int w, int h) {
    return "w * h";
  }
}
```

```
$ javac Area.java
Area.java:3: error: incompatible types:
String cannot be converted to int
    return "w * h";
           ^
1 error
```

Type checking a Lean program...

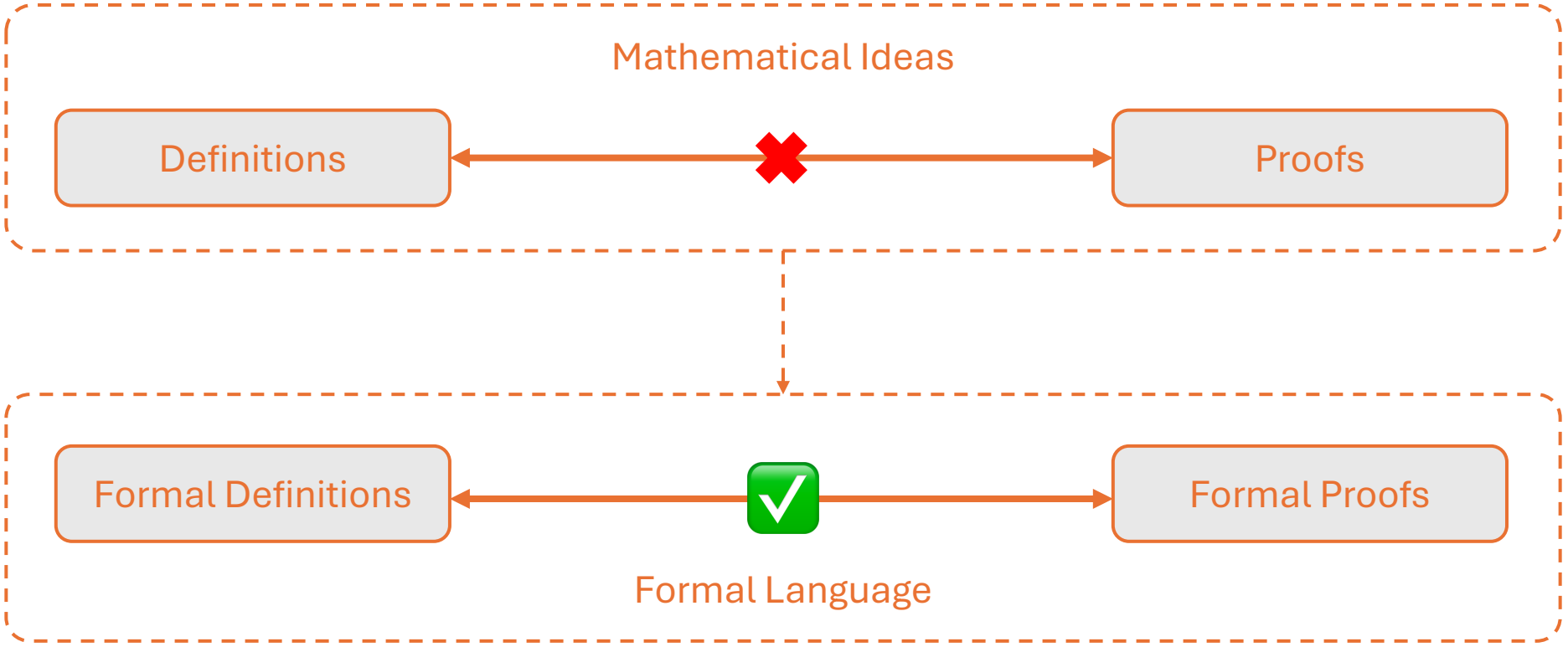
```
theorem zero_add (n : Nat) : 0 + n = n :=
  by
  induction n with
  | zero => rfl
  | succ k ih => rw [Nat.add_succ, ih]
```

```
$ lean P.lean
✓ no errors
```

```
theorem zero_add (n : Nat) : 0 + n = n :=
  by
  induction n with
  | zero => rfl
  | succ k ih => exact ih
```

```
$ lean F.lean
F.lean:5:17: error: Type mismatch
  ih
has type
  0 + k = k
but is expected to have type
  0 + (k + 1) = k + 1
```

Central Thesis



Central Thesis

1. Informal mathematics can be written as a statement in a formal language intended to carry the same mathematical meaning.
   - a statement in a theorem prover is a type

2. A theorem prover is that language plus a small kernel that mechanically checks the proof. If it compiles, the proof is valid.
   - relative to the axioms, the kernel, and the statement you actually wrote

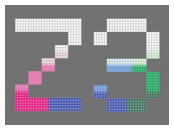
Formal Verification

Theorem Provers

Automated Theorem Provers (ATP)

Given — Formal Statement

Produce — Formal Proof, Certificate



CVC5

Interactive Theorem Provers (ITP)

Given — Formal Statement, Formal Proof

Produce — Certificate

LEAN

ROCQ



Theorem Provers

Automated Theorem Provers (ATP)

Given

Formal Statement

VERY HARD!!!

Produce

Certificate

CVC5

Interactive Theorem Provers (ITP)

Given

Formal Statement

Formal Proof

Produce

Certificate

LEAN

ROCQ



Theorem Provers

Automated Theorem Provers (ATP)

Given — Formal Statement

VERY HARD!!!

Produce — Certificate

CVC5

Interactive T

A. Where do we get the Formal Statement?

Given — Formal Statement, Formal Proof

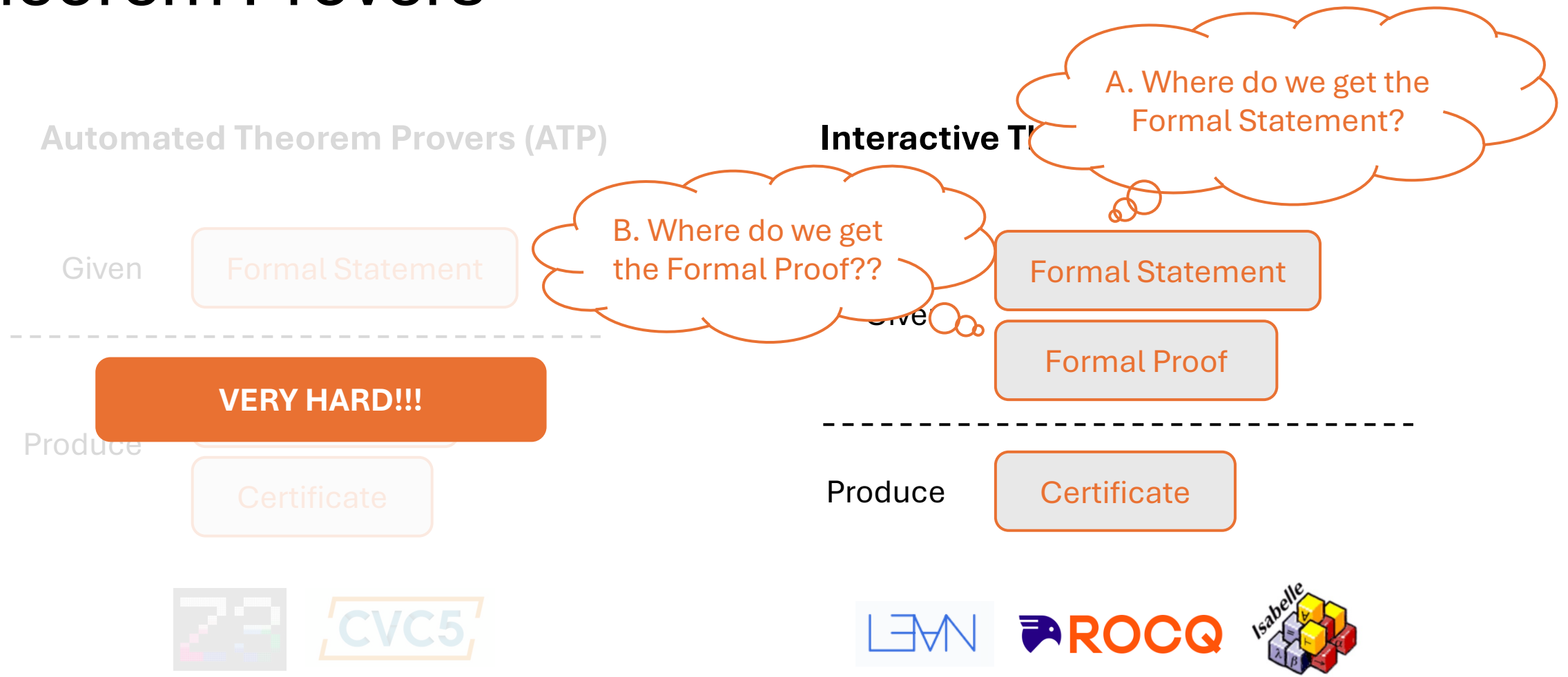
Produce — Certificate



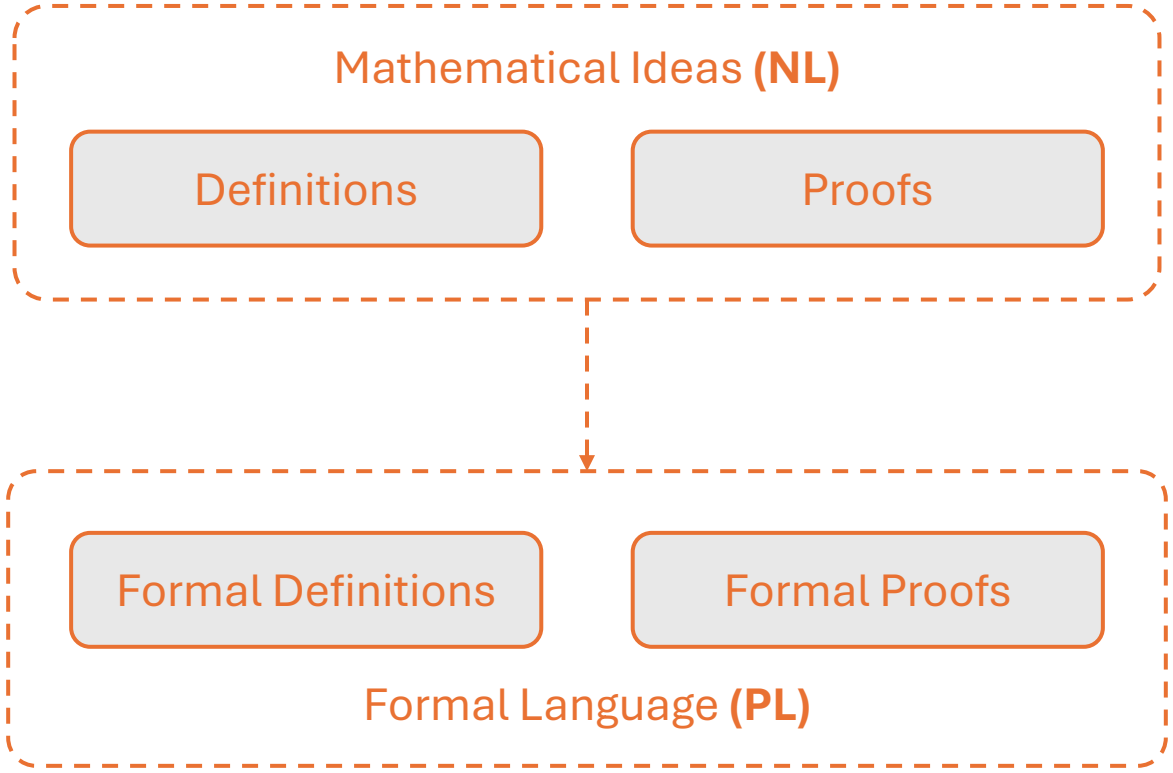




Theorem Provers



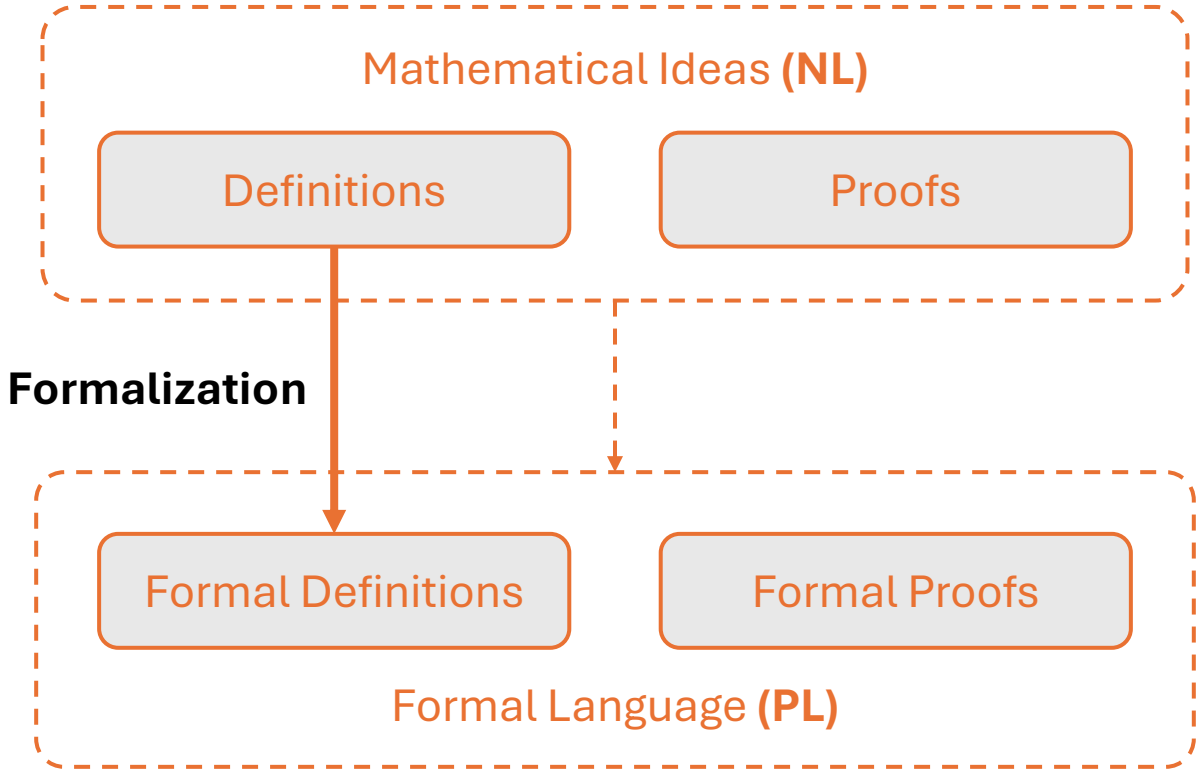
Problem Definition for Machine Programming



Natural Language (NL)

Programming Language (PL)

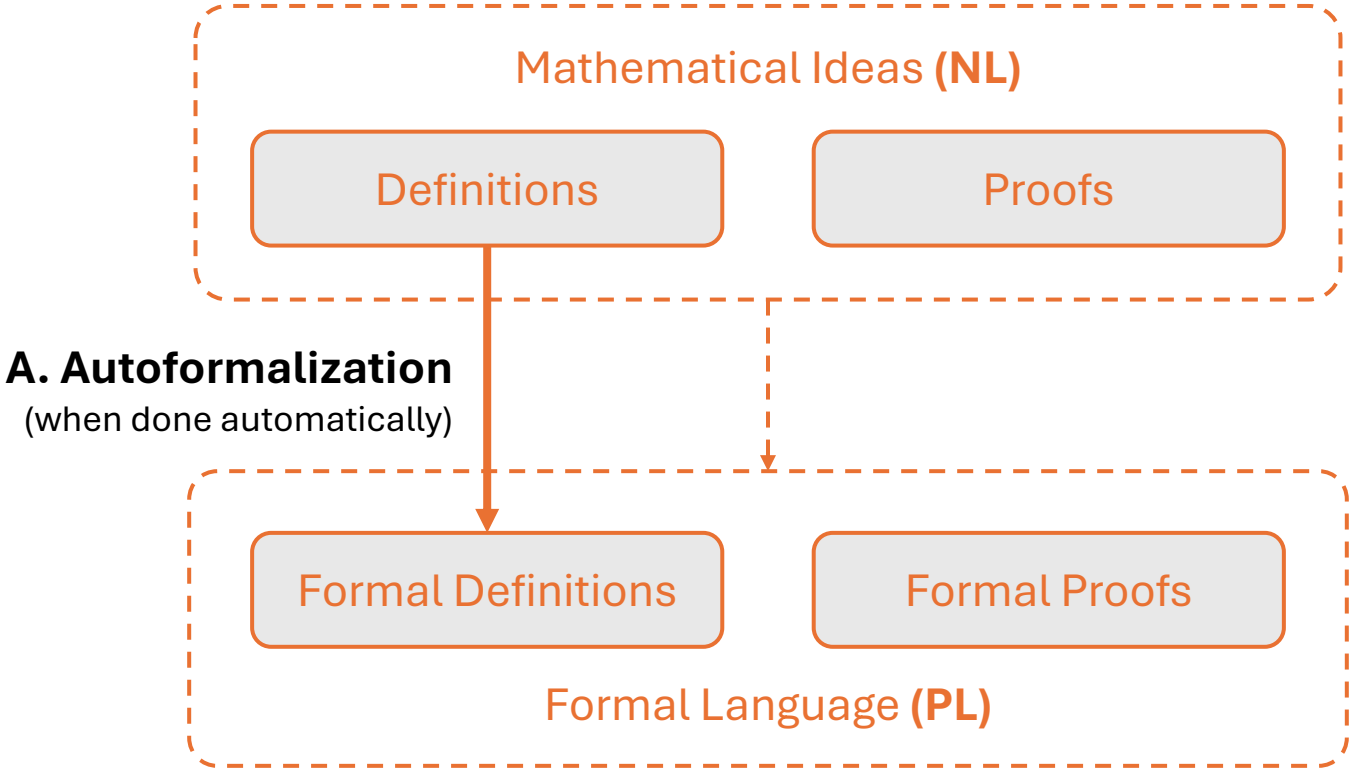
Problem Definition for Machine Programming



Natural Language (NL)

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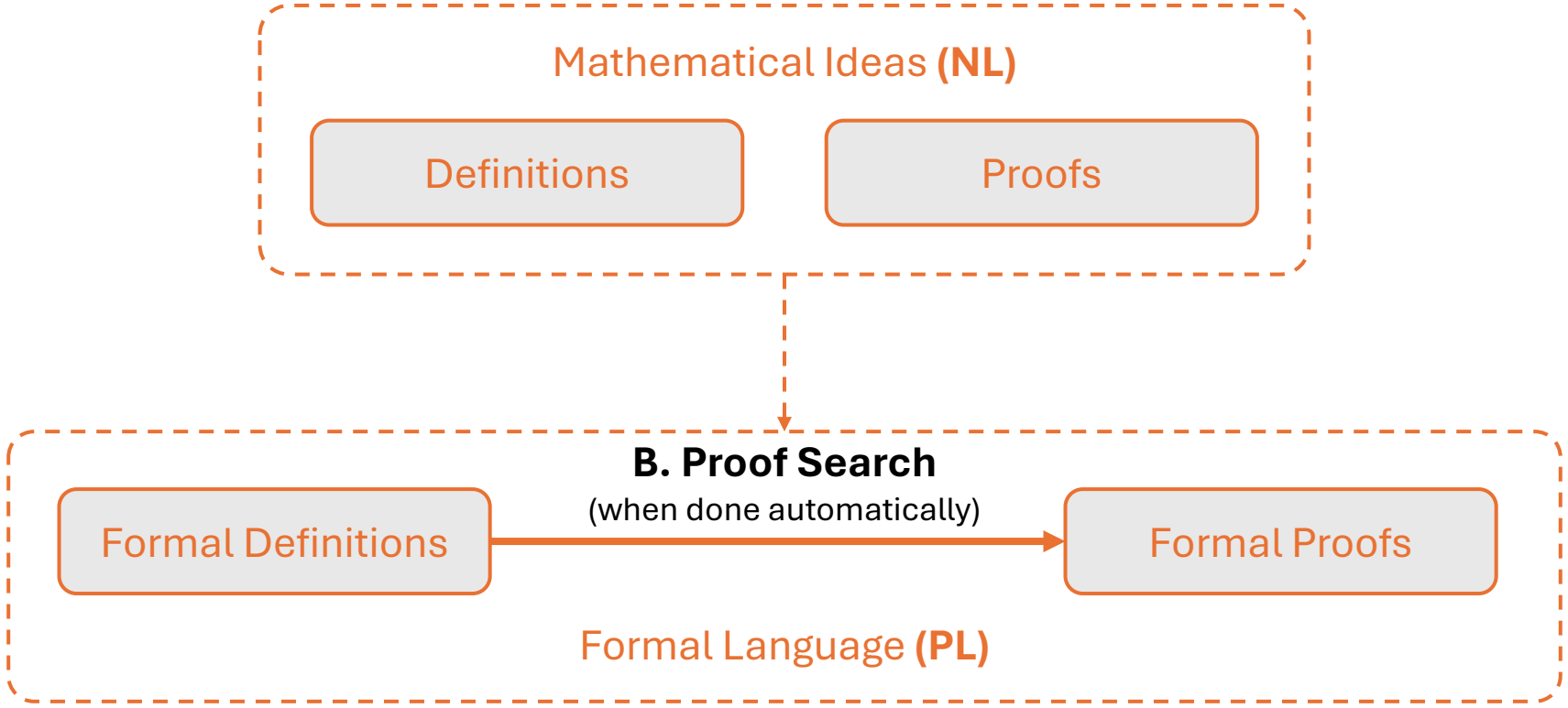
Problem Definition for Machine Programming



Natural Language (NL)

Programming Language (PL)

Problem Definition for Machine Programming



Natural Language (NL)

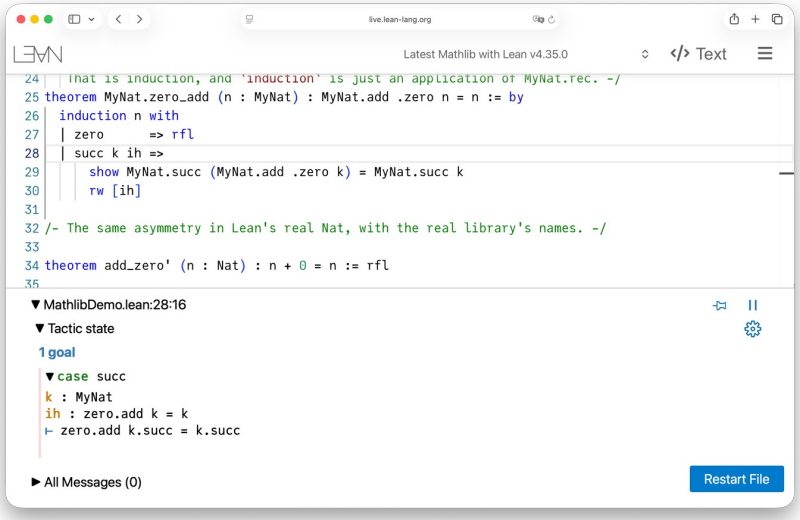
Programming Language (PL)

Agenda

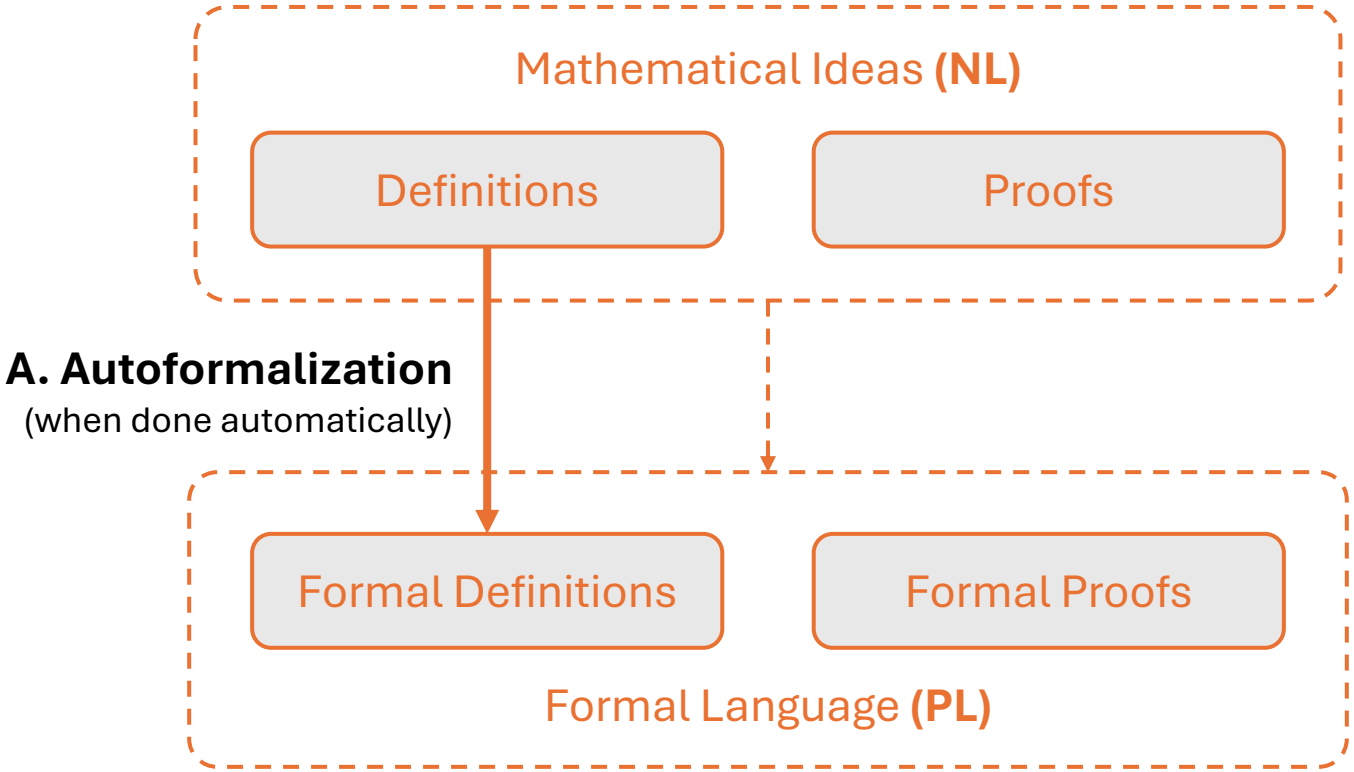
- Demo: agentic proof search
- Demo: verifying a simple program against specification
- Discussion: autoformalization and faithfulness

Demo Time

https://live.lean-lang.org



Autoformalization



Autoformalization Benchmarks

Published as a conference paper at ICLR 2022

MINIF2F: A CROSS-SYSTEM BENCHMARK FOR FORMAL OLYMPIAD-LEVEL MATHEMATICS

Kunhao Zheng
École Polytechnique
kunhao.zheng@polytechnique.edu

Jesse Michael Han
OpenAI
University of Pittsburgh
jessemichaelhan@openai.com

Stanislas Polu
OpenAI
spolu@openai.com

https://arxiv.org/abs/2109.00110

Autoformalization Ber

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Table 4: Problem 11 of 2000 AMC 12 is formalized with proof in different languages in miniF2F. The proof is optionally attached thus not part of the benchmark. The proof in Metamath is too long to be fully displayed.

Natural Language	Two non-zero real numbers, a and b, satisfy $ab = a - b$. Which of the following is a possible value of $\frac{a}{b} + \frac{b}{a} - ab$? (A) -2 (B) $\frac{-1}{2}$ (C) $\frac{1}{3}$ (D) $\frac{1}{2}$ (E) 2

Metamath	`${`
	`amc12-2000-p11.0 $e |- ( ph -> A e.  RR ) $.`
	`amc12-2000-p11.1 $e |- ( ph -> B e.  RR ) $.`
	`amc12-2000-p11.2 $e |- ( ph -> A =/= 0 ) $.`
	`amc12-2000-p11.3 $e |- ( ph -> B =/= 0 ) $.`
	`amc12-2000-p11.4 $e |- ( ph -> ( A x.  B ) =`
	`  ( A - B ) ) $.`
	`amc12-2000-p11 $p |- ( ph -> ( ( ( A / B ) +`
	`  ( B / A ) ) - ( A x.  B ) ) = 2 )`
	`$=`
	`( cdiv co caddc cmul cmin c2 cexp eqcomd ...  $.`
	`$}`

Lean	`theorem amc12_2000_p11`
	`  (a b :  ℝ)`
	`  (h₀ :  a ≠ 0 ∧ b ≠ 0)`
	`  (h₁ :  a * b = a - b) :`
	`  a / b + b / a - a * b = 2 :=`
	`begin`
	`  field_simp [h₀.1, h₀.2],`
	`  simp only [h₁, mul_comm, mul_sub],`
	`  ring,`
	`end`

Isabelle	`theorem amc12_2000_p11:`
	`  fixes a b::real`
	`  assumes "a \<noteq> 0" "b \<noteq> 0"`
	`  and "a * b = a - b"`
	`  shows "a / b + b / a - a * b = 2"`
	`  using assms`
	`  by (smt (verit, ccfv_threshold)`
	`  diff_divide_distrib`
	`  div_self divide_divide_times_eq`
	`  eq_divide_imp nonzero_mult_div_cancel_left)`
	`end`

https://arxiv.org/abs/2109.00110

Autoformalization Benchmarks

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MINIF2F: A CROSS-SYSTEM BENCHMARK FOR FORMAL OLYMPIAD-LEVEL MATHEM

Kunhao Zheng
École Polytechnique
kunhao.zheng@polytechnique.edu

Jesse M
OpenAI
Universi
jesser

Stanislas Polu
OpenAI
spolu@openai.com

PUTNAMBENCH: Evaluating Neural Theorem-Provers on the Putnam Mathematical Competition

George Tsoukalas UT Austin
Jasper Lee UT Austin
John Jennings UT Austin
Jimmy Xin UT Austin
Michelle Ding UT Austin
Michael Jennings UT Austin
Amitayush Thakur UT Austin
Swarat Chaudhuri UT Austin

https://arxiv.org/abs/2109.00110 https://arxiv.org/abs/2407.11214

Autoformalization Benchmarks

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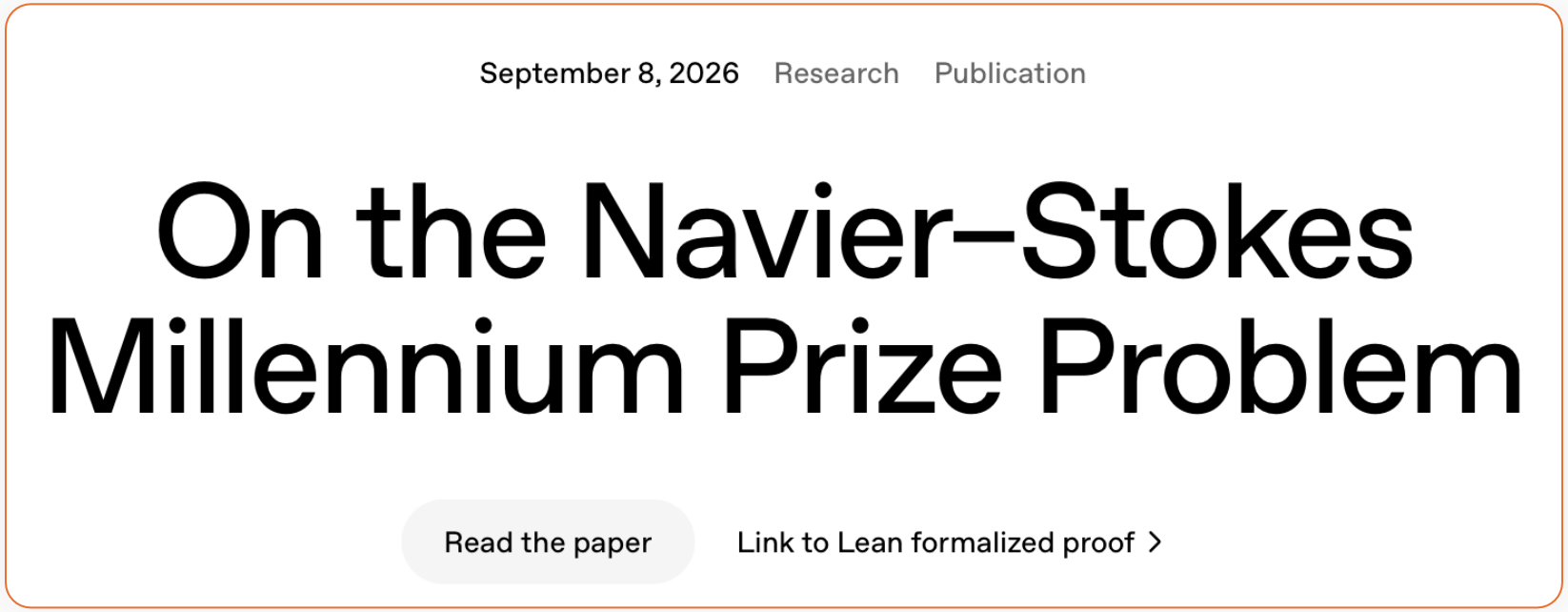
Amitayush Thakur
UT Austin

```
theorem putnam_1988_b1 :
∀ a ≥ 2, ∀ b ≥ 2, ∃ x y z : ℤ,
x > 0 ∧ y > 0 ∧ z > 0 ∧
a * b = x * y + x * z + y * z + 1 := by
    intro a ha b hb
    use a - 1, b - 1, 1
    constructor
    linarith
    constructor
    linarith
    constructor
    linarith
    ring
```

Figure 1: A formalization of Putnam 1988 B1 in Lean 4, which asserts that for all integers $a, b \geq 2$, there are positive integers x, y, z such that $ab = xy + xz + yz + 1$. The formal proof begins by introducing all relevant variables and hypotheses with `intro`, then indicating the choice of x, y, z with `use`, and afterwards proving all goals using the automated tactics `linarith` and `ring`. This proof was discovered through a few-shot invocation of GPT-4.

https://arxiv.org/abs/2109.00110 https://arxiv.org/abs/2407.11214

Discussion: Faithfulness in Autoformalization



https://openai.com/index/navier-stokes-solution/

Discussion: Faithfulness in Autoformalization

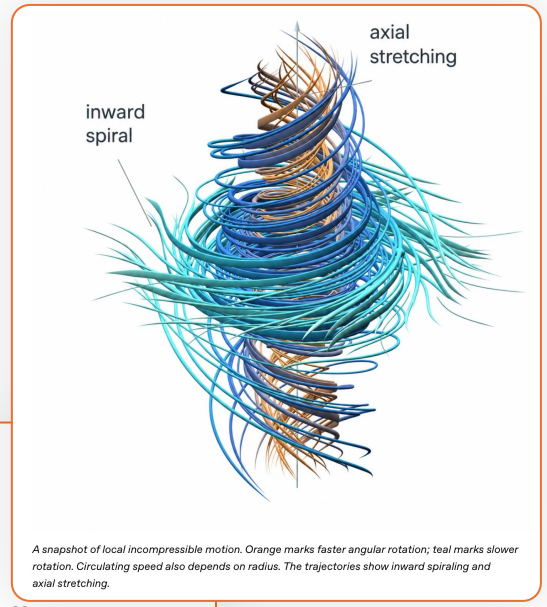
September 8, 2026 Research Publication

On the Navier–Stokes Millenni

Read

The result

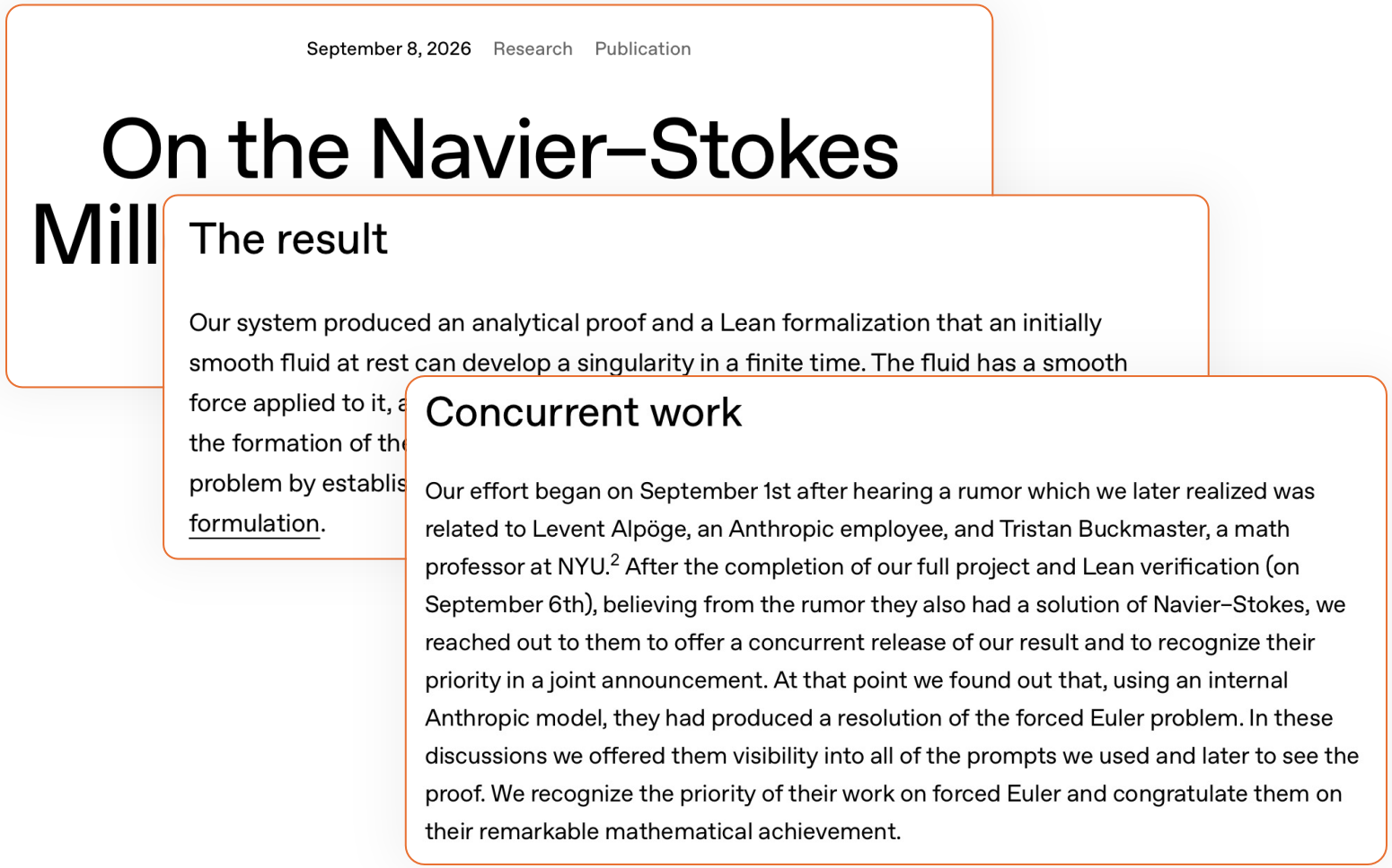
Our system produced an analytical proof and a Lean formalization that an initially smooth fluid at rest can develop a singularity in a finite time. The fluid has a smooth force applied to it, and its energy remains finite through the entire dynamics, from rest to the formation of the singularity. This resolves the Navier–Stokes Millennium Prize problem by establishing statement “C” (and also “D”) in the official Millennium Prize formulation.



A snapshot of local incompressible motion. Orange marks faster angular rotation; teal marks slower rotation. Circulating speed also depends on radius. The trajectories show inward spiraling and axial stretching.

https://openai.com/index/navier-stokes-solution/

Discussion: Faithfulness in Autoformalization



https://openai.com/index/navier-stokes-solution/

Discussion: Faithfulness in Autoformalization

September 8, 2026 Research Publication

On the Navier–Stokes Mill

The result

Our system produced an analytical proof and a Lean formalization that an initially smooth fluid at rest can develop a singularity in a finite time. The fluid has a smooth force applied to it, a
the formation of th
problem by establis
formulation.

Concurrent work

Our effort began on September 1st after hearing a rumor which we later realized was related to Levent Alpöge, an Anthropic employee, and Tristan Buckmaster, a math professor at NYU.[2] After the completion of our full project and Lean verification (on September 6th), believing from the rumor they also had a solution of Navier–Stokes, we reached out to them to offer a concurrent release of our result and to recognize their priority in a joint announcement. At that point we found out that, using an internal Anthropic model, they had produced a resolution of the forced Euler problem. In these discussions we offered them visibility into all of the prompts we used and later to see the proof. We recognize the priority of their work on forced Euler and congratulate them on their remarkable mathematical achievement.

Today, Levent Alpöge and I have made public three results: finite-time blowup with smooth forcing for incompressible porous media, for Boussinesq, and for 3d incompressible Euler.

We believe we also have blowup for hypo-dissipative Navier-Stokes. We are not releasing that paper today: unlike the above, the Lean verification has not yet finished. We do not yet have anything resembling a presentable writeup. I mention it because it is suggestive of a path to unforced Euler.

The program this fits into was not started by us nor was it proposed by a Large Language Model. The credit for the basic idea of this program goes to Diego Córdoba and Luis Martínez-Zoroa, who for several years have been exploring the construction of forced blow ups. We took their work as a starting point, using Large Language Models to push their program to completion.

Concretely, what Levent and I did was to take the Córdoba and Martínez-Zoroa program, which achieved blowup results with rough forcing, and, with a great deal of help from LLMs, push it to smooth forcing and to the incompressible Euler equations. The ideas making this line of attack possible are due to Córdoba and Martínez-Zoroa. Let me make plain what I have said to colleagues in private: in view of this body of work, I believe Luis Martínez-Zoroa deserves a Fields Medal.

My work with Levent has been a purely personal collaboration, free of any institutional agreements or official involvement by either of our employers. We used several LLMs throughout: Anthropic's Claude, OpenAI's Codex, especially with GPT-5.6 Sol and, more recently, Astra. The latter was only used for writeups and auditing our arguments.

For most of the past year progress was slow. We worked through the literature and upgraded various preliminary results, up to obtaining finite time blow up for the Incompressible Porous Media equation (with smooth forcing). This was until about a month ago, when we had real progress: on August 15th, we obtained the blow up results, with smooth forcing, for both Boussinesq and Euler. I can say the first LLM generated proof Levent sent me was the most horrendous I have ever read; we verified it on Lean on August 22nd. Since this point, we have been working around the clock to understand this proof and turn it into something readable.

There is another part of this story, and one that, honestly, I very much wish I did not have to be concerned with. I am not happy about the presentation quality in these papers. Ideally, we would have preferred to spend weeks turning the LLM generated proofs into something readable from the very first page. This level of care is what these problems and the community devoted to these problems deserves. The various Boussinesq and Euler write-ups in particular are much closer to what models produce under human direction than to a paper written by a person. The Euler writeup, in particular, can only be described as AI slop. I am sorry for this. The reasons are below, and they involve our being pressured by outside factors. I say this not as an excuse but as an explanation.

I had planned to say on announcing our work that the results are not the important thing. Rather the important thing is instead the significance that a mathematician and an LLM model can now do all this work in a month. The

1

https://cims.nyu.edu/~tristanb/statement.pdf

Discussion: Faithfulness in Autoformalization

FINITE TIME BLOWUP FOR NAVIER–STOKES

OPENAI

Abstract. For every positive viscosity, we construct a solution of the three-dimensional incompressible Navier–Stokes equations that starts from rest and develops unbounded velocity in finite time while maintaining uniformly bounded kinetic energy.

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1. Introduction

We consider the three-dimensional incompressible Navier–Stokes equations. A fundamental question about the nature of these equations is whether a solution starting from smooth data can develop unbounded velocity in finite time while its kinetic energy remains bounded. We construct such a flow with zero initial velocity and a smooth force compactly supported in space and time.

Theorem 1.1. *For every $\nu > 0$ there exist a force $f \in C_c^\infty(\mathbb{R}^3 \times (0,\infty); \mathbb{R}^3)$, a compact set $K \subset \mathbb{R}^3$, and smooth velocity and pressure fields u, p on $\mathbb{R}^3 \times [0,1)$ satisfying*

$$\partial_t u + (u \cdot \nabla)u - \nu\Delta u + \nabla p = f, \qquad \nabla \cdot u = 0, \qquad u(\cdot,0) = 0, \tag{1.1}$$

such that $\operatorname{supp} u(\cdot,t) \cup \operatorname{supp} p(\cdot,t) \subset K$ *for every* $0 \le t < 1$,

$$\sup_{0\le t<1} \|u(t)\|_{L^2(\mathbb{R}^3)} < \infty, \qquad \limsup_{t\uparrow 1} \|u(t)\|_{L^\infty(\mathbb{R}^3)} = \infty.$$

Consequently, there is no smooth solution (u,P) on $\mathbb{R}^3 \times [0,\infty)$ with the same force and initial datum whose kinetic energy is uniformly bounded $\sup_{t\ge 0} \frac{1}{2}\int_{\mathbb{R}^3} |u(x,t)|^2\,dx < \infty$.

This establishes alternative (C) in the Millennium problem statement for Navier–Stokes as stated by Fefferman in [13]. Compact support also yields the corresponding construction on $\mathbb{T}^3 = \mathbb{R}^3/\mathbb{Z}^3$, establishing alternative (D) in [13]; see Corollary 10.6.

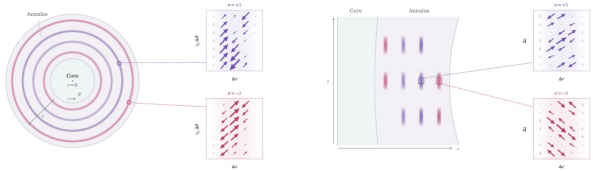


Figure 2. Pulse geometry and perturbation velocities. *Left:* At fixed height, pulse envelopes occupy complete rings; the close-ups show radial and azimuthal perturbation velocities. *Right:* A radial–axial section, where each localized patch is a cross-section of a complete ring; the close-ups show radial and axial perturbation velocities. Colors distinguish the two families. The specific pulse arrangement shown is schematic; active locations, spacing, and aspect ratios are illustrative.

rates satisfy

$$\frac{|u_r|}{\ell_r} = O(\tau^{-1}), \qquad \frac{|u_z|}{\ell_z} \asymp \tau^{-1}, \qquad \frac{\nu}{\ell_r^2} \asymp \tau^{-1}.$$

Thus radial diffusion and transport by the radial and axial flow remain in the leading balance as the core contracts. Axial diffusion is weaker: the ratio of axial to radial diffusion rates is

$$\frac{\nu/\ell_z^2}{\nu/\ell_r^2} = \frac{\ell_r^2}{\ell_z^2} \asymp \tau^{2h} \longrightarrow 0.$$

The core must also provide enough shear at its outer edge to amplify the pulses placed there. Away from the middle plane, axial flow carries angular momentum from more rapidly rotating layers, supporting a steep radial decrease in angular velocity. With exact reflection symmetry, this transport would vanish at $z = 0$. Symmetry would also impose $u_z(r,0,t) = 0$, leaving no radial shear of the axial velocity to compensate. We therefore choose a slightly asymmetric axial profile, with a small upward bias and nonzero velocity at $z = 0$. This separates the layer where rotational amplification is weak from the layer where axial shear vanishes. The radial shear of axial velocity then supplies the additional amplification needed near the middle plane; farther away, the rotational mechanism already suffices.

2.2. Annulus and pulses.

At the edge of the core, there is an annular region where the inner vortex joins onto an exterior flow. The background satisfies its leading momentum equations inside the core, but this transition leaves an imbalance in the annulus. The external force needed to sustain the background becomes unbounded as $t \uparrow 1$, so it cannot serve as the smooth external force required by the construction.

We therefore perturb the velocity so that the fluid's own motion supplies the missing momentum transport. Through the nonlinear advection term, the perturbation produces additional momentum fluxes whose spatial variation acts as an internal force on the background. We require this contribution to cancel the singular part of the background residual, leaving only a remainder that extends smoothly through $t = 1$. These internal forces redistribute momentum within the fluid.

the pulse-end radius defined before (A.10). Beyond it, $U = M = 0$ by Proposition A.4 and (C.17); thus $\mathcal{A}_X(U) = M/X = 0$, and (4.7) gives $V_0 = 0$. This proves the remaining assertion of part (v).

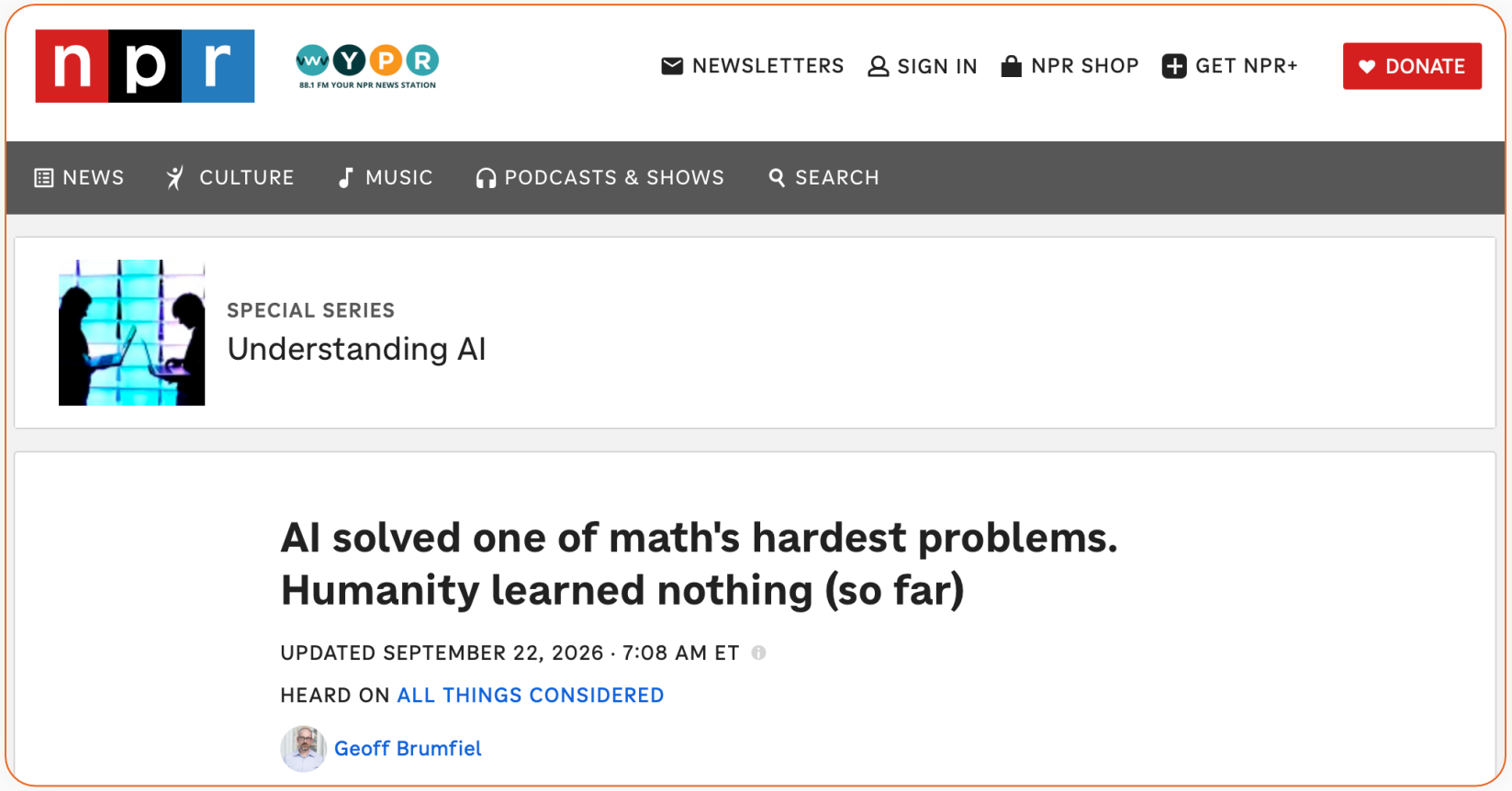
Finally, the third and fourth reserved intervals listed after (A.9) are the intervals $\mathcal{I}_{\mathrm{pos}}, \mathcal{I}_{\mathrm{mean}}$ in part (vi). They lie in (X_a, X_v), and their listed order gives $\sup \mathcal{I}_{\mathrm{pos}} < \inf \mathcal{I}_{\mathrm{mean}}$. Proposition C.2 leaves them unchanged. The input profiles there satisfy $U = 0$ and $E = K(\eta)X^{-1/2-\lambda}$, with $K(\eta) = c_{\mathrm{patch}}(1+\eta^2)^{-1}$ for a fixed positive constant c_{patch}, by Proposition A.4. Thus they retain (4.30), completing part (vi). The finite frequency N was fixed in Proposition C.2 before the physical scale q and all later bands and correction stages, so the constants have the independence asserted at the end of Theorem 4.6. □

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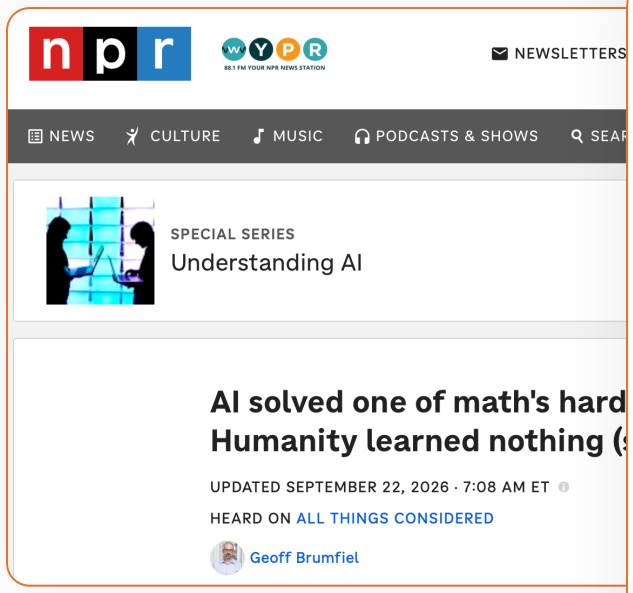
https://github.com/openai/NavierStokesAndEuler

Discussion: Faithfulness in Autoformalization



https://www.npr.org/2026/09/22/nx-s1-5968588/openai-navier-stokes-problem-mathematicians-learn-little

Discussion: Faithfulness in Autoformalization



When OpenAI announced its AI had solved one of the world's toughest math problems earlier this month, its tone was celebratory:

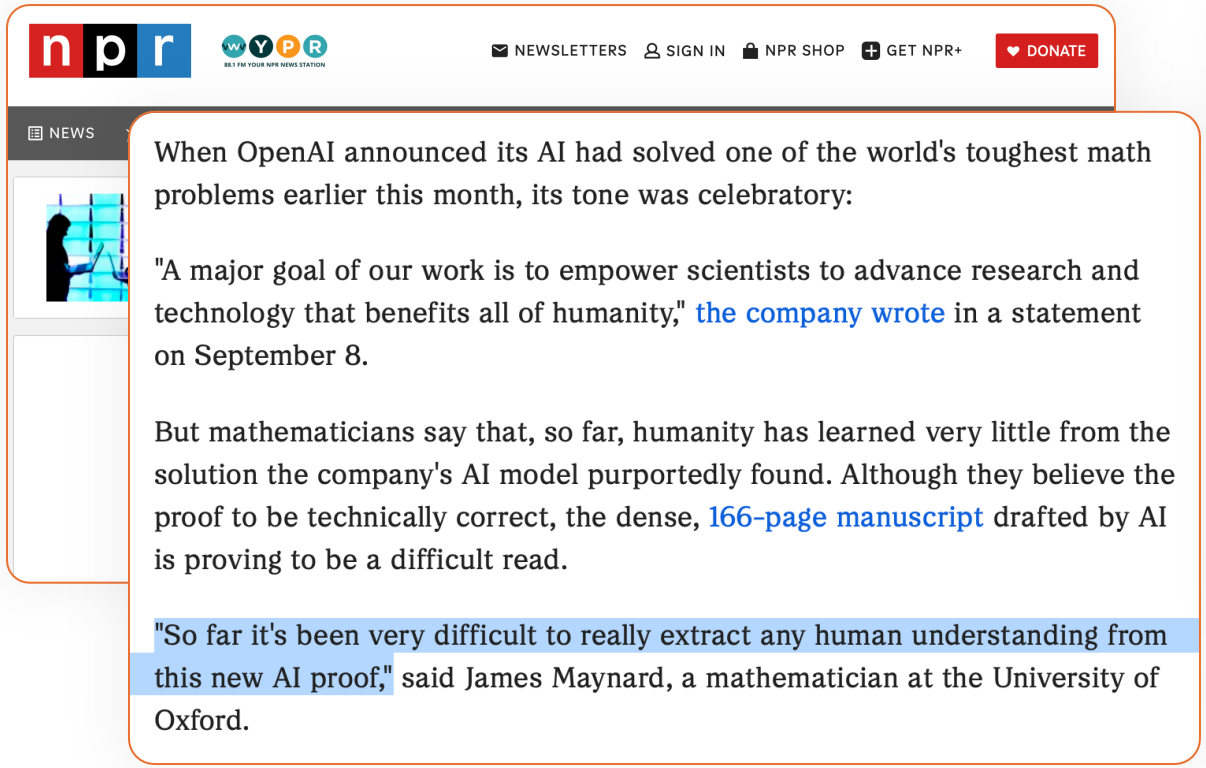
"A major goal of our work is to empower scientists to advance research and technology that benefits all of humanity," the company wrote in a statement on September 8.

But mathematicians say that, so far, humanity has learned very little from the solution the company's AI model purportedly found. Although they believe the proof to be technically correct, the dense, 166-page manuscript drafted by AI is proving to be a difficult read.

"So far it's been very difficult to really extract any human understanding from this new AI proof," said James Maynard, a mathematician at the University of Oxford.

https://www.npr.org/2026/09/22/nx-s1-5968588/openai-navier-stokes-problem-mathematicians-learn-little

Discussion: Faithfulness in Autoformalization



OpenAI Has Claimed To Have Produced A Proof To Resolve The Navier–Stokes $1 Million Millennium Prize Problem: Reports

SEPTEMBER 8, 2026 OFFICECHAI TEAM

AI might have produced its biggest breakthrough yet.

OpenAI has reportedly produced a proof cracking one of the most famous unsolved problems in mathematics — the Navier–Stokes equations — using an enormous fleet of AI agents, according to reports from Axios and Wired.

Fields Medalist Terence Tao, meanwhile, has called Buckmaster and Alpöge's own AI-assisted, Lean-verified proofs "a remarkable achievement" and suggested the work could genuinely help solve Navier–Stokes — while separately warning that a world where AI produces unreadable proofs the community can't parse would be "a disaster" for mathematics.

https://officechai.com/ai/openai-has-claimed-to-have-produced-a-proof-to-resolve-the-navier-stokes-1-million-millennium-prize-problem-reports/

Discussion: Faithfulness in Autoformalization

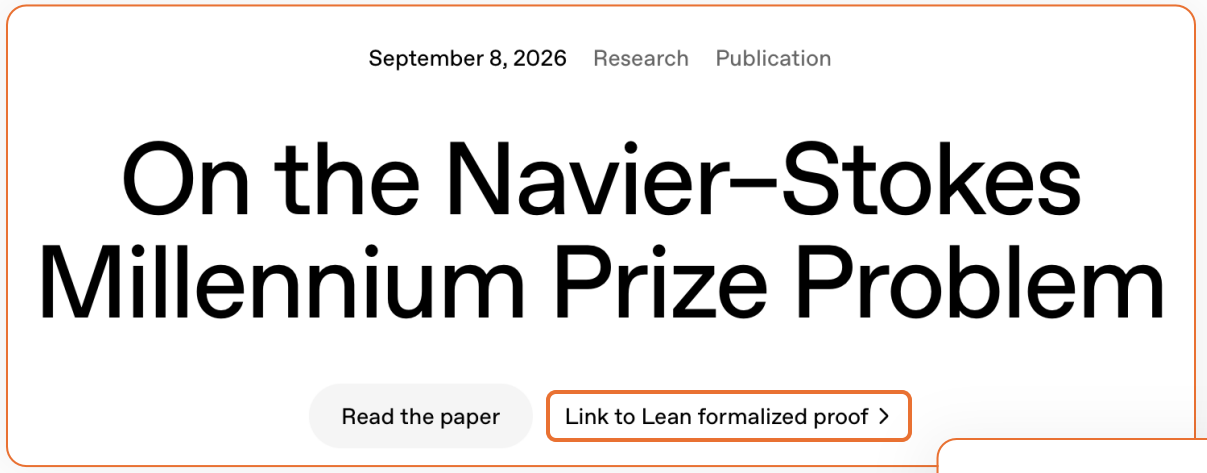
- **James Maynard (Oxford, Fields Medalist)** — told NPR "So far it's been very difficult to really extract any human understanding from this new AI proof". NPR also notes the dense, 166-page AI-drafted manuscript is proving a difficult read and is taking weeks to untangle, though few mathematicians believe it's wrong. AI solved one of math's hardest problems. Humanity learned nothing (so far)

- **Javier Gómez-Serrano (Brown)** — "The paper is not written for humans." He said it could help the field after serious rewriting, but that as of now it doesn't teach much. Futurism

- **Terence Tao (UCLA)** — warned that a world where AI produces unreadable proofs the community can't parse would be a disaster for mathematics (general warning, not a specific "I can't read it" claim). Officechai

- **General framing** — Science News: since the paper is 166 pages long, mathematicians are still digesting it; Futurism: mathematicians agree the results are valid, though unpacking them will take many more weeks. Science News / Futurism

Discussion: Faithfulness in Autoformalization

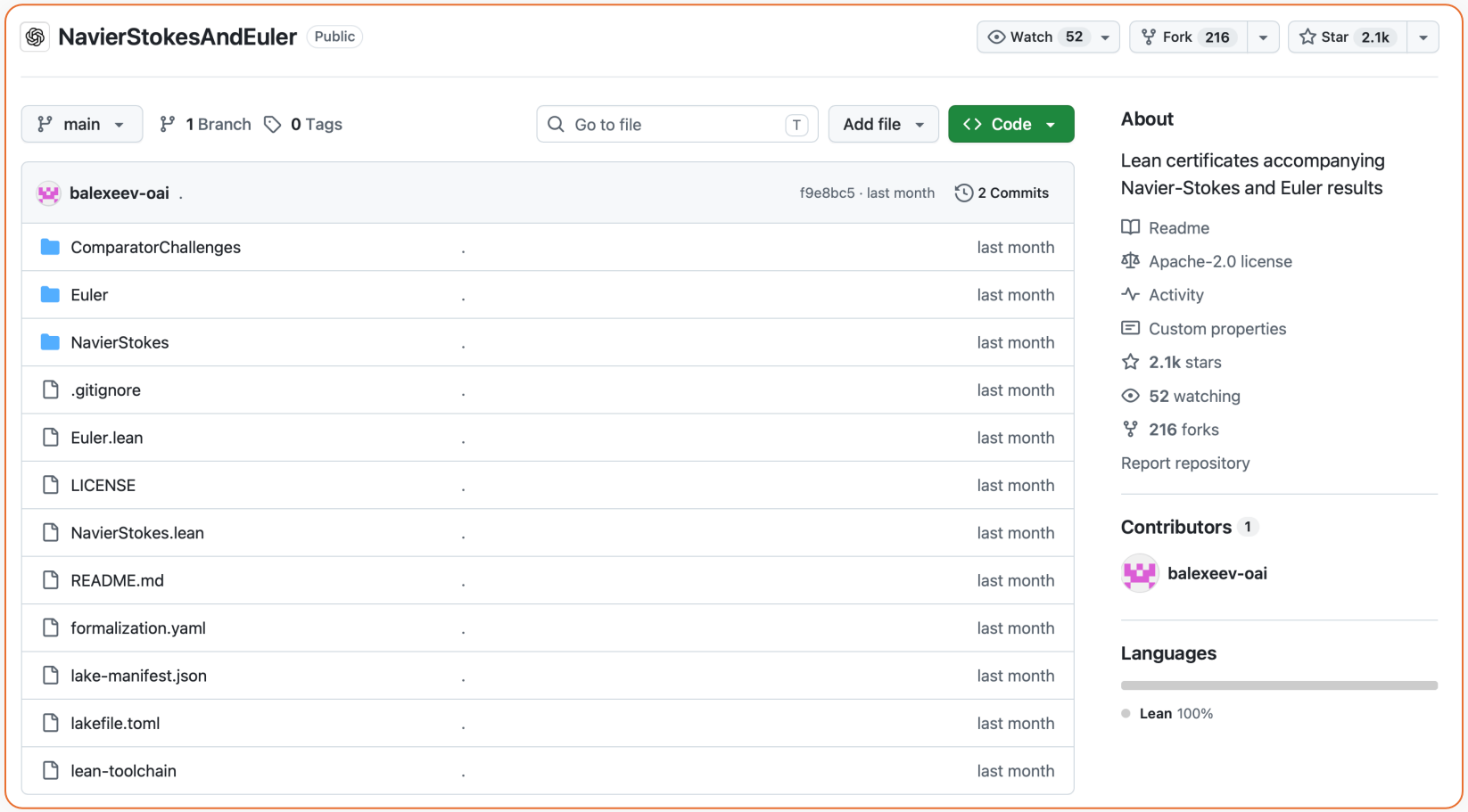


The result

Our system produced an analytical proof and a Lean formalization that an initially smooth fluid at rest can develop a singularity in a finite time. The fluid has a smooth force applied to it, and its energy remains finite through the entire dynamics, from rest to the formation of the singularity. This resolves the Navier–Stokes Millennium Prize problem by establishing statement “C” (and also “D”) in the official Millennium Prize formulation.

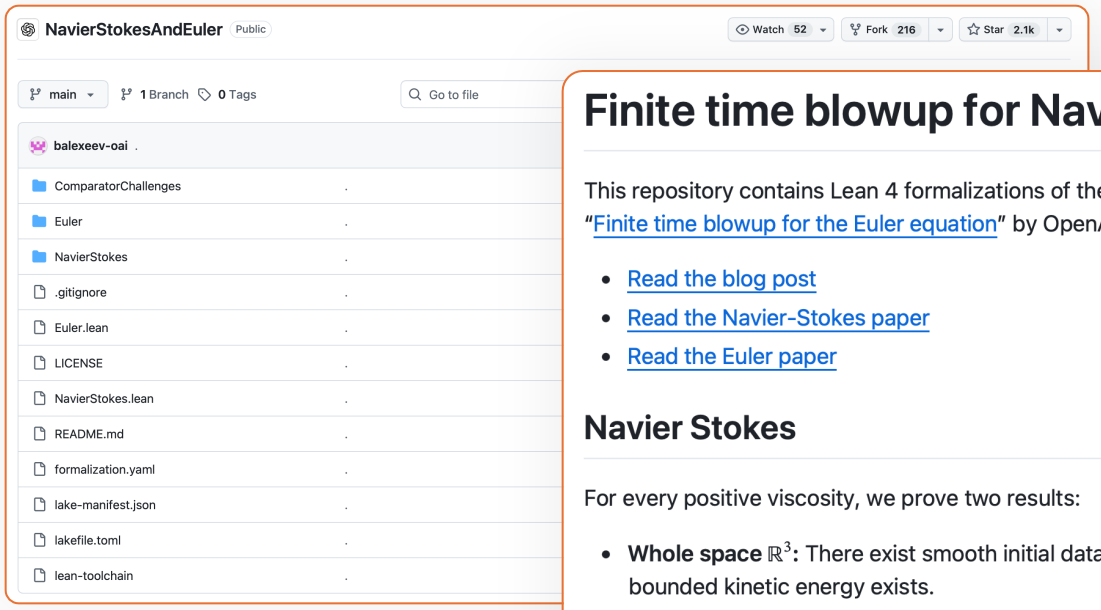
https://openai.com/index/navier-stokes-solution/

Discussion: Faithfulness in Autoformalization



https://github.com/openai/NavierStokesAndEuler

Discussion: Faithfulness in Autoformalization



Finite time blowup for Navier–Stokes and Euler equations

This repository contains Lean 4 formalizations of the results presented in "Finite time blowup for Navier–Stokes" and "Finite time blowup for the Euler equation" by OpenAI.

- Read the blog post
- Read the Navier-Stokes paper
- Read the Euler paper

Navier Stokes

For every positive viscosity, we prove two results:

- **Whole space $\mathbb{R}^3$:** There exist smooth initial data and forcing for which no global smooth solution with uniformly bounded kinetic energy exists.
- **Periodic torus $\mathbb{R}^3 / \mathbb{Z}^3$:** There exist smooth periodic initial data and forcing for which no global smooth solution exists.

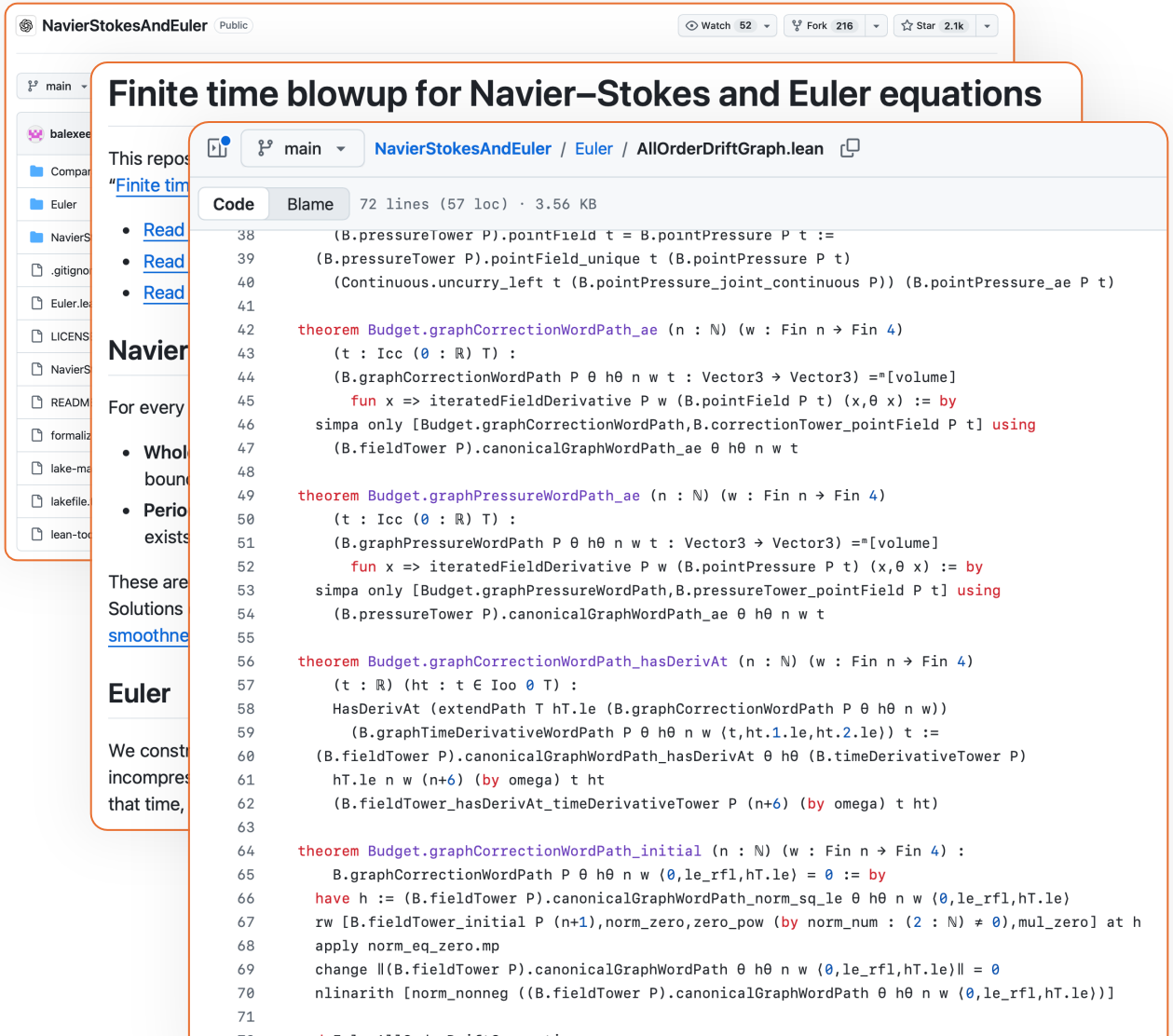
These are alternatives (C) "Breakdown of Navier–Stokes solutions on $\mathbb{R}^3$" and (D) "Breakdown of Navier–Stokes Solutions on $\mathbb{R}^3/\mathbb{Z}^3$" in the Clay Mathematics Institute's official problem description of the Navier–Stokes existence and smoothness Millennium Prize Problem.

Euler

We construct smooth, compactly supported, divergence-free initial velocity on $\mathbb{R}^3$ whose solution to the unforced incompressible Euler equations develops a singularity in finite time. The velocity's C^1 norm becomes unbounded near that time, and the time integral of the vorticity's L^∞ norm diverges.

https://github.com/openai/NavierStokesAndEuler

Discussion: Faithfulness in Autoformalization



https://github.com/openai/NavierStokesAndEuler

Discussion: Faithfulness in Autoformalization

arXiv

Mathematics > Analysis of PDEs

[Submitted on 6 Oct 2026]

Navier–Stokes lost in translation: Why Lean verification of AI autoformalisation does not guarantee correct natural language proofs

Alexander Bastounis, Fabian Circelli, Anders C. Hansen

Autoformalisation is increasingly used to verify mathematical texts, including those generated by AI, as in OpenAI's announced proof of blow-up of solutions to the Navier–Stokes equations. In this process, an AI system translates the text from a natural language (NL) into a formal language such as Lean. Once this translation is done, the argument expressed in the formal language can easily be mechanically verified. The purpose of this article is to demonstrate why this process may offer no confidence in the original NL argument, owing to the various difficulties in performing the translation semantically faithfully. In particular, we highlight that the problem of resolving ambiguities in mathematical NL text, which is necessary in order to provide semantically faithful translation, is arbitrarily high up in the Solvability Complexity Index (SCI) hierarchy/arithmetical hierarchy (the SCI $= \infty$). Hence, informally, providing semantically faithful AI autoformalisation is harder than any computational problem including the Halting problem (which has SCI $= 1$). To demonstrate the effect of this result we provide several examples of AI mistranslations of NL statements and proofs into Lean in practice, resulting in mismatches between NL proofs and their Lean `verifications'. These include OpenAI's announced Navier–Stokes proof. In particular, we show that the formalised Lean proof does not correspond to the NL proof of blow-up of solutions to the Navier–Stokes equations.

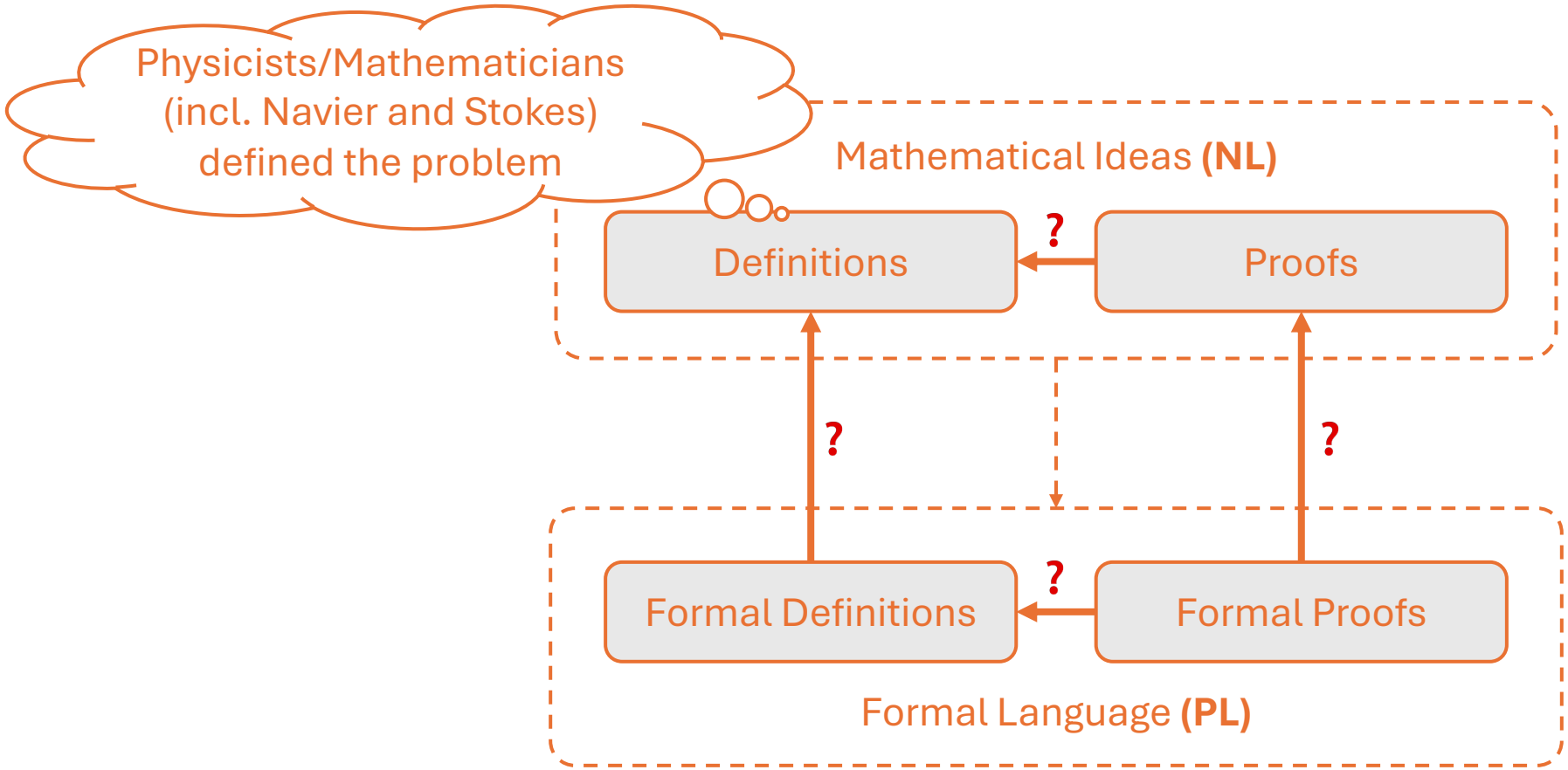
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Subjects:	**Analysis of PDEs (math.AP)**; Artificial Intelligence (cs.AI); Logic (math.LO)
MSC classes:	35Q30, 03Dxx (primary) and 68V20, 68Txx, 03B65 (secondary)
Cite as:	arXiv:2610.08144 **[math.AP]**
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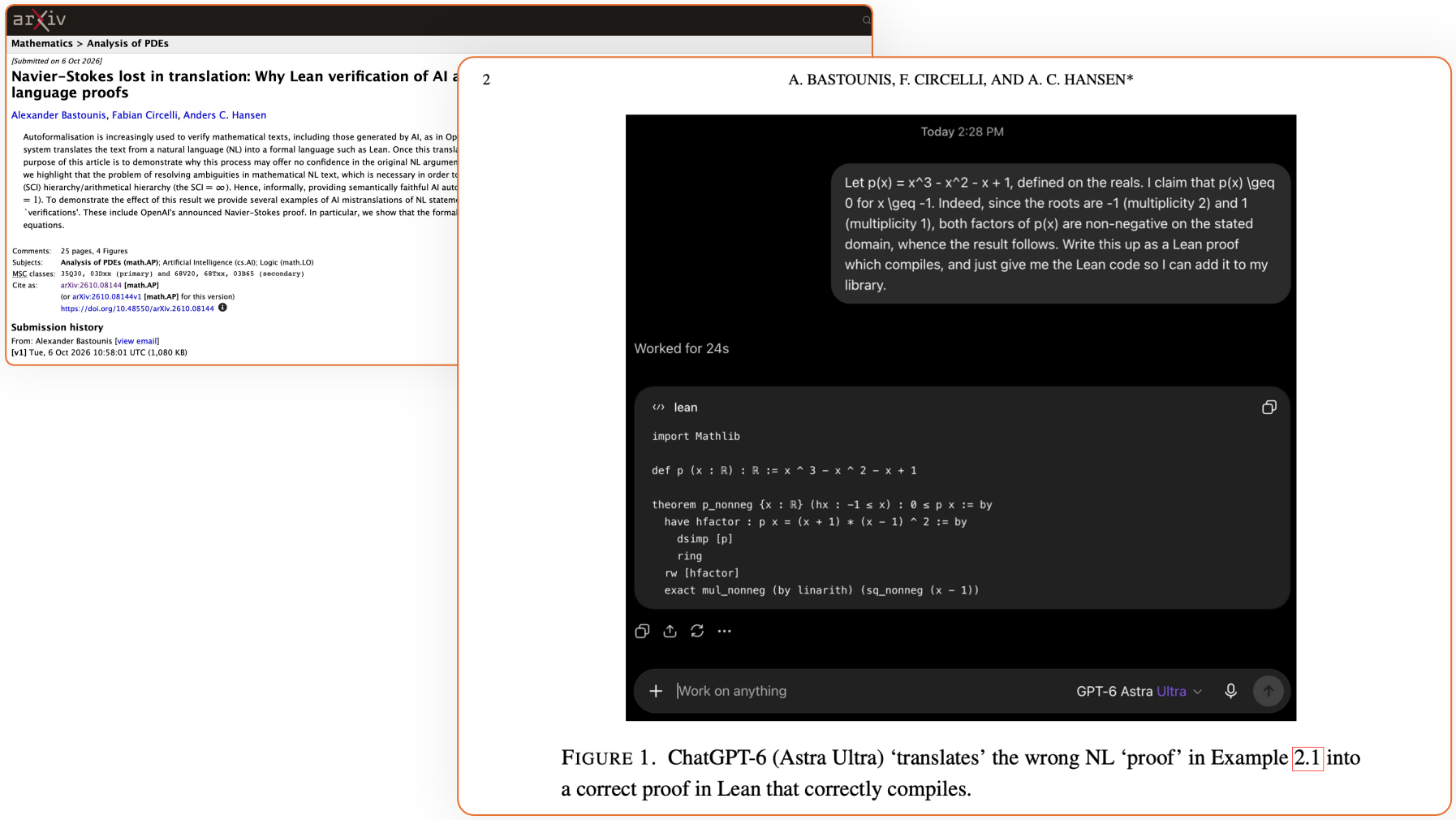
From: Alexander Bastounis [view email]
[v1] Tue, 6 Oct 2026 10:58:01 UTC (1,080 KB)

https://arxiv.org/pdf/2610.08144

What is happening for Navier-Stokes?



Discussion: Faithfulness in Autoformalization



arXiv

Mathematics > Analysis of PDEs

[Submitted on 6 Oct 2026]

Navier–Stokes lost in translation: Why Lean verification of AI a… language proofs

Alexander Bastounis, Fabian Circelli, Anders C. Hansen

Autoformalisation is increasingly used to verify mathematical texts, including those generated by AI, as in Op… system translates the text from a natural language (NL) into a formal language such as Lean. Once this transl… purpose of this article is to demonstrate why this process may offer no confidence in the original NL argumen… we highlight that the problem of resolving ambiguities in mathematical NL text, which is necessary in order to… (SCI) hierarchy/arithmetical hierarchy (the SCI = ∞). Hence, informally, providing semantically faithful AI auto… = 1). To demonstrate the effect of this result we provide several examples of AI mistranslations of NL stateme… `verifications'. These include OpenAI's announced Navier–Stokes proof. In particular, we show that the formal… equations.

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https://doi.org/10.48550/arXiv.2610.08144

Submission history

From: Alexander Bastounis [view email]
[v1] Tue, 6 Oct 2026 10:58:01 UTC (1,080 KB)

Today 2:28 PM

Let p(x) = x^3 - x^2 - x + 1, defined on the reals. I claim that p(x) \geq 0 for x \geq -1. Indeed, since the roots are -1 (multiplicity 2) and 1 (multiplicity 1), both factors of p(x) are non-negative on the stated domain, whence the result follows. Write this up as a Lean proof which compiles, and just give me the Lean code so I can add it to my library.

Worked for 24s

lean

```
import Mathlib

def p (x : ℝ) : ℝ := x ^ 3 - x ^ 2 - x + 1

theorem p_nonneg {x : ℝ} (hx : -1 ≤ x) : 0 ≤ p x := by
  have hfactor : p x = (x + 1) * (x - 1) ^ 2 := by
    dsimp [p]
    ring
  rw [hfactor]
  exact mul_nonneg (by linarith) (sq_nonneg (x - 1))
```

Work on anything GPT-6 Astra Ultra

FIGURE 1. ChatGPT-6 (Astra Ultra) 'translates' the wrong NL 'proof' in Example 2.1 into a correct proof in Lean that correctly compiles.

https://arxiv.org/abs/2610.08144

Discussion: Faithfulness in Autoformalization

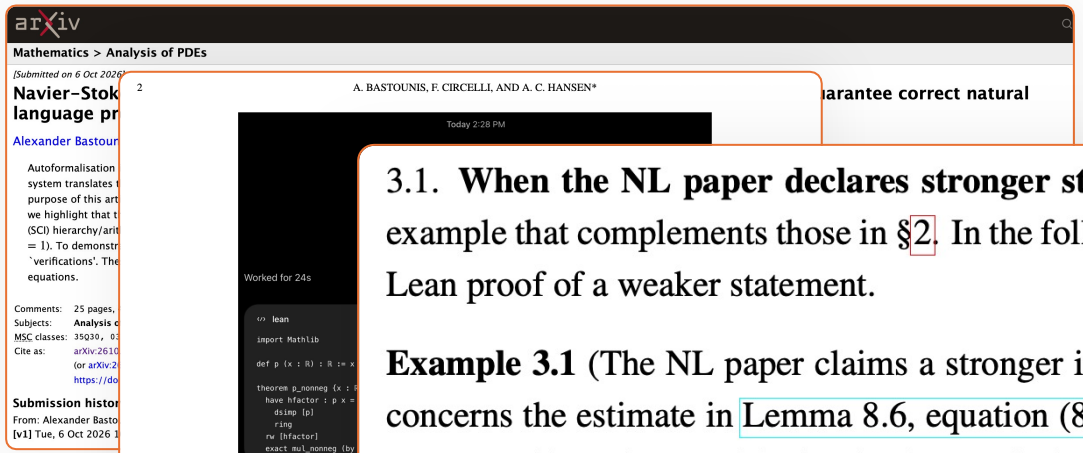


FIGURE 1. ChatGPT-6 (A
a correct proof in Lean tha

3.1. **When the NL paper declares stronger statements than what Lean proves.** We commence with an example that complements those in §2. In the following example the AI autoformalisation results in a different Lean proof of a weaker statement.

Example 3.1 (The NL paper claims a stronger inverse estimate than the cited Lean formalisation)**.** Figure 3 concerns the estimate in Lemma 8.6, equation (8.19), of OpenAI's *Finite time blowup for Navier–Stokes*. We compare this estimate with the cited Lean declarations at commit `f9e8bc5` (click to see the documentation on Github). The issue is a change in the order of the input derivatives required to bound the output derivatives. In particular, we have the following discrepancy. With m a nonnegative integer, the comparison in Figure 3 is

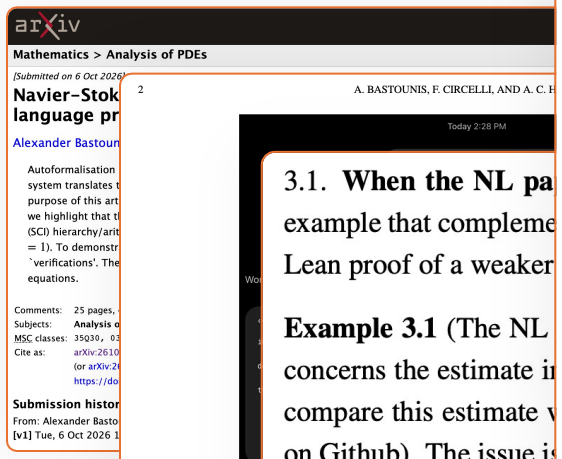
$$\text{NL paper, equation (8.19):} \qquad \|N^{-1}F\|_{C^m_y} \leqslant C_m \|F\|_{C^{m+4}_y}, \tag{3.1}$$

$$\text{Lean Estimate:} \qquad \|N^{-1}F\|_{C^m_y} \leqslant K_m \|F\|_{C^{m+5}_y}. \tag{3.2}$$

Summary: The Lean code produced by OpenAI often has stronger conditions on bounds on the derivatives of various functions than the corresponding results in the NL paper.

https://arxiv.org/abs/2610.08144

Discussion



The proof of Lemma 8.6 opens with the following passage (source paper, page 96):

> "The divisor bound (6.7) gives $|v_t \cdot k|^{-1} \leqslant C(1+|k|)$. Four more derivatives of F than the requested output leave a summable $(1+|k|)^{-3}$ bound on the Fourier series in two dimensions. This proves (8.19), smoothness, and uniqueness after fixing the zero Fourier coefficient. The multiplier commutes with every slow coefficient derivative and preserves reality."

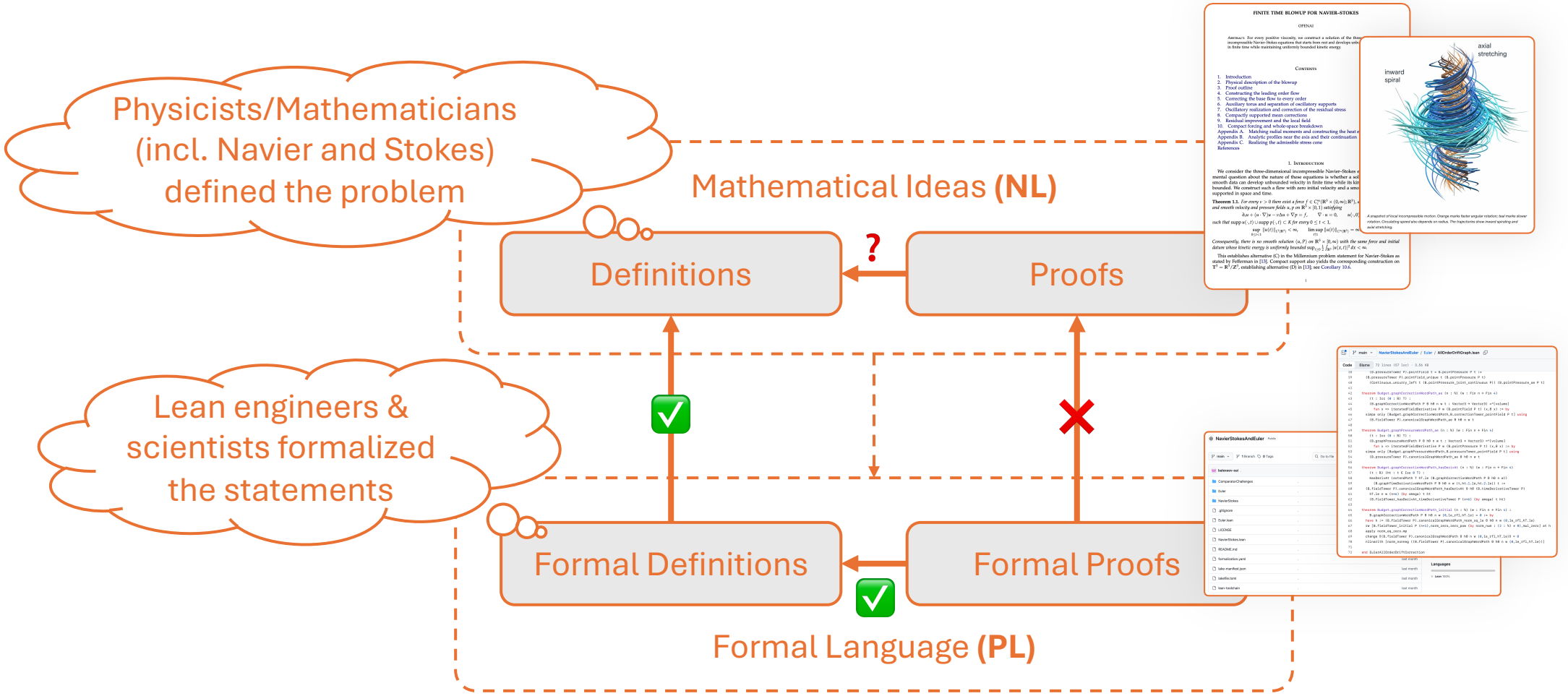
The related Lean codes – on the other hand – all use five additional derivatives, not four. A key example is the following Lean theorem, which appears to bear the closest resemblance to (8.19):

```
omit [FiniteDimensional ℝ P] in
theorem norm_derivativeWord_inverse_le (d : Direction) {f : Source P}
    (hf : ContDiff ℝ ∞ f) (hp : Periodic f) (w : List Bool) (p : P) (Y : Plane) {C : ℝ}
    (hzero : ∀ x ∈ Icc (0 : ℝ) 1, ∀ y ∈ Icc (0 : ℝ) 1, ‖f (p, (x, y))‖ ⩽ C)
    (hfirst : ∀ x ∈ Icc (0 : ℝ) 1, ∀ y ∈ Icc (0 : ℝ) 1,
      ‖SmoothFourierData.xJet (w.length + 5) (slice f p) (x, y)‖ ⩽ C)
    (hsecond : ∀ x ∈ Icc (0 : ℝ) 1, ∀ y ∈ Icc (0 : ℝ) 1,
      ‖SmoothFourierData.xJet (w.length + 5)
        (SmoothFourierData.swapFunction (slice f p)) (x, y)‖ ⩽ C) :
    ‖derivativeWord w (slice (inverse d f) p) Y‖ ⩽
      ParametricTorusInverse.mixedLossConstant w.length * C := by
  have hbound := SmoothFourierData.coefficient_seminorm_bound (slice_smooth hf p) (hp p)
    (w.length + 1) hzero (by simpa only [Nat.add_assoc] using hfirst)
    (by simpa only [Nat.add_assoc] using hsecond)
  calc
    _ ⩽ ((6 * ‖omega⁻¹‖) * ‖omega‖ ^ w.length) *
        coeffSeminorm (w.length + 1) (coefficient f p) :=
      inverse_derivativeWord_bound d
        (SmoothFourierData.rapid_coefficient (slice_smooth hf p) (hp p)) w Y
    _ ⩽ ((6 * ‖omega⁻¹‖) * ‖omega‖ ^ w.length) *
        ((3 ^ ((w.length + 1) + 4) * C) * ∑' k : Frequency, (weight k ^ 4)⁻¹) :=
      mul_le_mul_of_nonneg_left hbound (by positivity)
    _ = _ := by simp only [ParametricTorusInverse.mixedLossConstant, Nat.add_assoc]; ring
```

Source (GitHub permalinks): `openai/NavierStokesAndEuler`, commit `f9e8bc5`.
File: `NavierStokes/SmoothFamilyTorusInverse.lean`.
Declaration: `norm_derivativeWord_inverse_le`, lines 1059–1080.
Red has been added to highlight aspects relevant to this example. The displayed Lean code is otherwise unchanged.

https://arxiv.org/abs/2610.08144

What is happening for Navier-Stokes?



What is happening for Navier-Stokes?

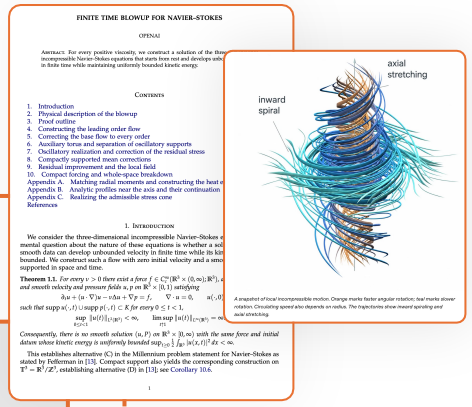
Physicists/Mathematicians (incl. Navier and Stokes) defined the problem

Mathematical Ideas **(NL)**

Definitions

?

Proofs

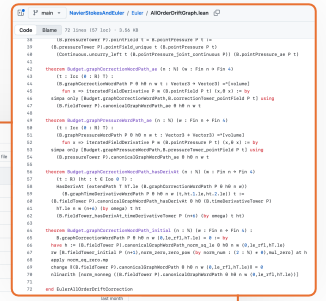


Lean engineers & scientists formalized the statements

✅



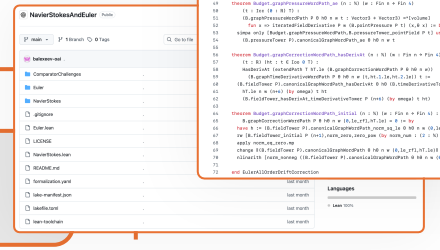
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Formal Definitions

✅

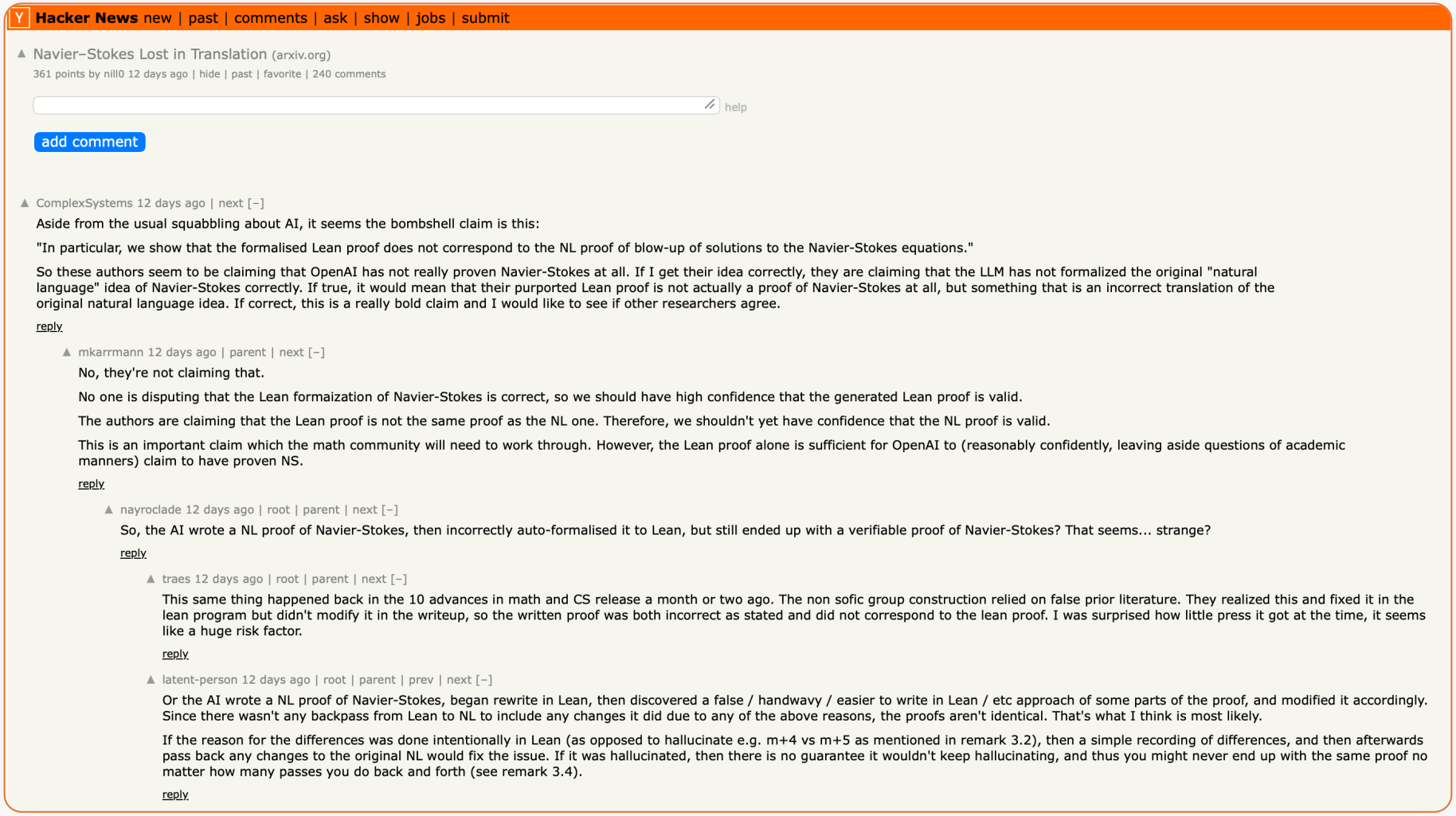
Formal Proofs



Formal Language **(PL)**



Discussion: Faithfulness in Autoformalization



Hacker News new | past | comments | ask | show | jobs | submit

▲ Navier–Stokes Lost in Translation (arxiv.org)

361 points by nill0 12 days ago | hide | past | favorite | 240 comments

help

add comment

▲ ComplexSystems 12 days ago | next [–]

Aside from the usual squabbling about AI, it seems the bombshell claim is this:

"In particular, we show that the formalised Lean proof does not correspond to the NL proof of blow-up of solutions to the Navier-Stokes equations."

So these authors seem to be claiming that OpenAI has not really proven Navier-Stokes at all. If I get their idea correctly, they are claiming that the LLM has not formalized the original "natural language" idea of Navier-Stokes correctly. If true, it would mean that their purported Lean proof is not actually a proof of Navier-Stokes at all, but something that is an incorrect translation of the original natural language idea. If correct, this is a really bold claim and I would like to see if other researchers agree.

reply

▲ mkarrmann 12 days ago | parent | next [–]

No, they're not claiming that.

No one is disputing that the Lean formaization of Navier-Stokes is correct, so we should have high confidence that the generated Lean proof is valid.

The authors are claiming that the Lean proof is not the same proof as the NL one. Therefore, we shouldn't yet have confidence that the NL proof is valid.

This is an important claim which the math community will need to work through. However, the Lean proof alone is sufficient for OpenAI to (reasonably confidently, leaving aside questions of academic manners) claim to have proven NS.

reply

▲ nayroclade 12 days ago | root | parent | next [–]

So, the AI wrote a NL proof of Navier-Stokes, then incorrectly auto-formalised it to Lean, but still ended up with a verifiable proof of Navier-Stokes? That seems... strange?

reply

▲ traes 12 days ago | root | parent | next [–]

This same thing happened back in the 10 advances in math and CS release a month or two ago. The non sofic group construction relied on false prior literature. They realized this and fixed it in the lean program but didn't modify it in the writeup, so the written proof was both incorrect as stated and did not correspond to the lean proof. I was surprised how little press it got at the time, it seems like a huge risk factor.

reply

▲ latent-person 12 days ago | root | parent | prev | next [–]

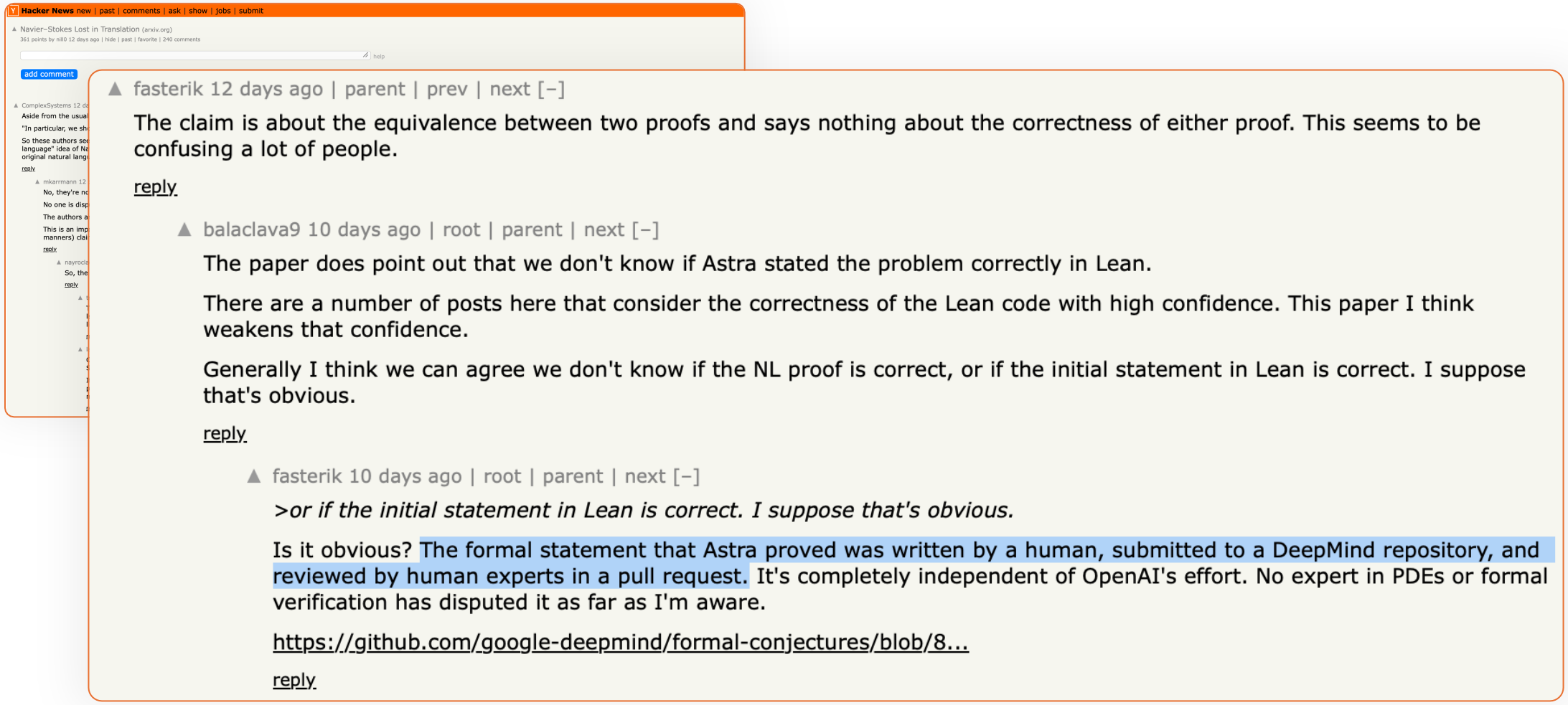
Or the AI wrote a NL proof of Navier-Stokes, began rewrite in Lean, then discovered a false / handwavy / easier to write in Lean / etc approach of some parts of the proof, and modified it accordingly. Since there wasn't any backpass from Lean to NL to include any changes it did due to any of the above reasons, the proofs aren't identical. That's what I think is most likely.

If the reason for the differences was done intentionally in Lean (as opposed to hallucinate e.g. m+4 vs m+5 as mentioned in remark 3.2), then a simple recording of differences, and then afterwards pass back any changes to the original NL would fix the issue. If it was hallucinated, then there is no guarantee it wouldn't keep hallucinating, and thus you might never end up with the same proof no matter how many passes you do back and forth (see remark 3.4).

reply

https://news.ycombinator.com/item?id=49994145

Discussion: Faithfulness in Autoformalization



https://news.ycombinator.com/item?id=49994145

What is happening for Navier-Stokes?

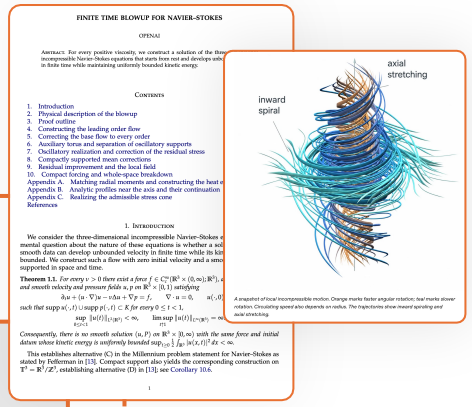
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Definitions

?

Proofs

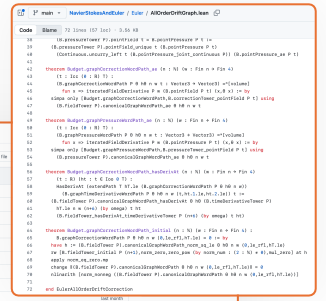


Lean engineers & scientists formalized the statements

✅



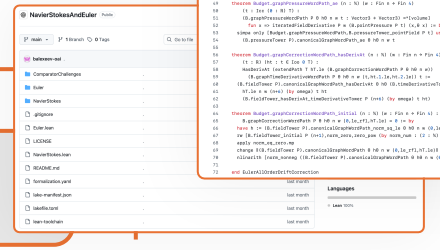
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Formal Definitions

✅

Formal Proofs



Formal Language **(PL)**



What is happening for Autoformalization?

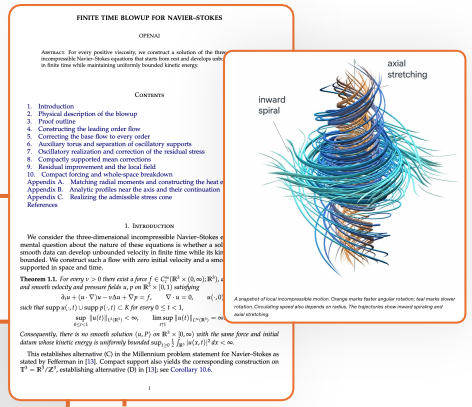
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?

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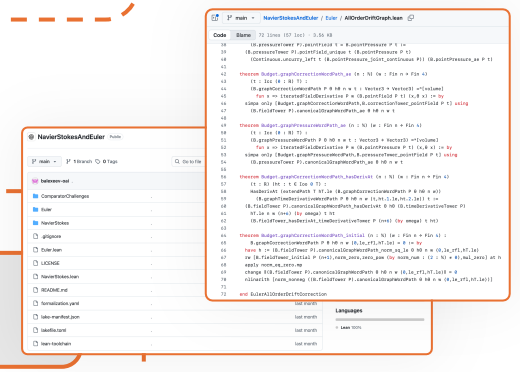
AI Generated...

?

?

Formal Definitions

Formal Proofs



✅

Formal Language **(PL)**

Assessing Faithfulness

- Does the formalization represent the same mathematical concepts in the original problem?

- Alt.: Does the formal program possess the same semantics as the informal mathematical definition?

- **Alt.: Is the formal program FAITHFUL to the informal definition?**

arXiv:2606.26525v1 [cs.LG] 25 Jun 2026

Theory-Scale Auto-Formalization of Logics for Computer Science

Yuming Feng
Johns Hopkins University
yfeng97@cs.jhu.edu

Frederick Pu
University of Toronto
frederick.pu@mail.utoronto.ca

One An
University of Pennsylvania
onean@sas.upenn.edu

Osbert Bastani
University of Pennsylvania
obastani@seas.upenn.edu

Li Zhang
Drexel University
harry.Zhang@drexel.edu

Jiani Huang
University of Pennsylvania
jianih@seas.upenn.edu

Xujie Si
University of Toronto
six@cs.toronto.edu

Ziyang Li
Johns Hopkins University
ziyang@cs.jhu.edu

Abstract

Auto-formalization is critical for scalable formal verification, but existing progress largely focuses on isolated statements, while *theory-scale* auto-formalization, which coherently translates hundreds of interdependent definitions, lemmas, and theorems, remains open due to challenges in consistency, faithfulness, scalability, and correctness. In this paper, we introduce *LCS-Bench*, a stand-alone, theory-scale benchmark based on *Logics for Computer Science*. LCS-Bench is built through a novel semi-automated agentic pipeline that leverages concept graphs, formal signature planning, issue tracking, sorry-filling with counter-example search, complemented by faithfulness review from human experts. The resulting artifact covers 327 textbook items, over 4,076 Lean declarations, and more than 85K lines of Lean code. The dataset supports broad evaluation through a data engine that automatically derives five tracks of evaluation benchmarks, measuring different aspects of auto-formalization and theorem-proving capabilities. We also introduce a novel evaluation protocol featuring definitional equivalence checkers, enabling more fine-grained and faithful assessment. Through extensive evaluation on 14 models, we demonstrate that (1) LCS-Bench is of high quality, consistent, and faithful; (2) the benchmark is challenging, with state-of-the-art models achieving only 20.1% on auto-formalization tasks; and (3) our analysis reveals key findings regarding theory-scale auto-formalization and suggests promising directions for future work.

1 Introduction

Auto-formalization, the translation of informal mathematics and logic into machine-verifiable statements and proofs, has become a central problem at the intersection of AI and formal methods [1–4]. Despite recent progress, *theory-scale* auto-formalization remains largely unsolved: real mathematical theories require hundreds of interdependent definitions, lemmas, and theorems to be formalized as a coherent whole. This setting raises fundamental challenges in maintaining global consistency, preserving faithfulness to the source text, and producing artifacts that are not only syntactically valid but also formally provable. Progress is further limited by the lack of tailored benchmarks and evaluation protocols, as many existing evaluations rely heavily on mature libraries such as Mathlib [5], obscuring the difficulty of building theories from first principles.

Preprint.

arXiv:2608.15432v1 [cs.AI] 15 Aug 2026

Does the Proof Prove It That Way? Faithful Formalization of Elements Proofs

Tadd Mao[1], **Tianjun Zhong**[2], **Dhruva Arekar**[2], **Yuming Feng**[2], **One An**[3], **Jiani Huang**[3], **Xujie Si**[1]*, **Ziyang Li**[2*†]
[1]University of Toronto
[2]Johns Hopkins University
[3]University of Pennsylvania
ziyang@cs.jhu.edu

Abstract

In formal verification, both the autoformalization of statements and automated proof search have been studied extensively. While automated proof search can produce a formal proof that compiles, the generated proof does not necessarily reflect how the natural-language argument arrives at its conclusion—a property we refer to as *faithfulness*. With faithfully formalized proofs, one can check the reasoning behind a human- or AI-written argument, and assist mathematicians in formalizing their proof sketches. However, it is particularly challenging due to misalignment of formal proof tactics and natural language reasoning. In this work, we rigorously describe a set of five necessary conditions a faithful formal proof must satisfy, and introduce Pistis, an agentic, oracle-guided proof search that produces formal Lean proofs that satisfy them. At its core is a novel faithfulness-preserving divide-and-conquer search, which we name OrderDecompose, that tracks citation dependencies and blocks unfaithful shortcuts, paired with a refutation search, that surfaces gaps and errors in the natural language proof source. OrderDecompose completes proofs that baselines cannot close even within a 12-hour budget, and its artifacts compile over 33× as fast as prior work's. We apply Pistis on the first three books of Euclid's *Elements*, producing high-quality artifacts containing faithful formal proofs. Under a blinded human study and an LLM-as-a-judge protocol on rigorous rubrics, Pistis-generated proofs are favored over prior works—2.89× and 5.2× as often by human reviewers and the LLM judge, respectively. It further uncovers gaps in Euclid's proofs and their translation, and can accept or refute natural language proofs written by humans or AI, demonstrating that faithful formalization is useful as a proof-checking tool.

1 Introduction

> We are not trying to meet some abstract production quota of definitions, theorems and proofs. The measure of our success is whether what we do enables people to understand and think more clearly and effectively about mathematics.
>
> On Proof and Progress in Mathematics (Thurston 1994)

Autoformalization translates natural-language (NL) mathematics into the formal language of a proof assistant such as Lean or Rocq, where it can be checked mechanically (Wu et al. 2022; de Moura and Ullrich 2021). Most work targets the formalization of *statements* (Li et al. 2024; Azerbayev et al. 2023), yet the formalization of *proofs* is often left to automated proof search, which hunts for *a* sequence of proof tactics that closes the goal. As a result, proof *faithfulness*, whether the formal proof reflects how the NL argument reaches its conclusion (Avigad, Dean, and Mumma 2009; Cabral et al. 2025), is largely overlooked.

Faithfulness is not a stylistic nicety; it is what makes a formal proof useful beyond a yes or no verdict. A faithful formalization lets us *check correctness at the level of the argument*, certifying that the reasoning as written in NL is valid. When the argument has a hole, a faithful attempt *localizes* it rather than silently papering over it with proof automation, letting us *validate or refute* a given NL proof. It helps *mathematicians* turn informal sketches into formal artifacts that still resemble what they wrote, and it keeps those artifacts *legible* for study and teaching, since each NL reasoning step maps directly to a formal step. For a proof to contribute to the broader field, legibility is only the starting point; the community must also recognize, digest, and build upon its argument. A human author can convey the insights behind a proof, while current AI tools remain black boxes (Tao 2026).

Achieving faithfulness is hard for three reasons. First, the target is intrinsically misaligned with the source: a proof assistant like Lean or Rocq advances by *tactics*, which transform proof states, whereas an NL proof advances by *sentences* of mathematical reasoning. Second, NL proofs may omit steps a human finds obvious but the formal system explicitly demands, so a faithful proof must *fill these gaps* without altering mathematical proof structures. Third, the field lacks a precise, checkable notion of what proof faithfulness *is* (Murphy et al. 2024; Cabral et al. 2025): prior evaluations lean on coarse proxies such as LLM-as-a-judge semantic matching (Lu et al. 2024; Poiroux et al. 2025; Yu et al. 2025), which cannot serve as a certification.

In this work we introduce Pistis[1], a framework that formalizes an NL proof into a faithful Lean proof, which we demonstrate in Euclidean geometry. Our key insight is that faithfulness becomes tractable once the proof is *segmented*: rather than searching for a single proof term that closes the whole theorem, we split the NL proof into structural formal

*These authors contributed equally.
†Corresponding author.

[1]From the Greek *pistis* (πίστις), meaning "faith" or "trust."

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Aligning Faithfulness for Theory-Scale Autoformalization

Yuming Feng[1] **Tadd Mao**[2] **Dhruva Arekar**[1] **Tianjun Zhong**[1]
Jiani Huang[3] **Xujie Si**[2] **Ziyang Li**[1]
[1]Johns Hopkins University [2]University of Toronto [3]University of Pennsylvania
yfeng97@cs.jhu.edu, tadd.mao@mail.utoronto.ca, darekar1@jhu.edu,
tzhong4@jhu.edu, jianih@seas.upenn.edu, six@cs.toronto.edu, ziyang@cs.jhu.edu

Abstract

Autoformalization (AF) requires more than compilation: formal statements must preserve the meaning of their informal sources, a property known as *faithfulness*. An emerging frontier is the automated evaluation of this property, yet most existing methods operate at the statement level and produce whole-item verdicts, failing to reveal how individual expressions correspond or which contextual interpretations support the judgment. This omission is consequential because AF is inherently theory-scale: target statements derive their meaning from surrounding mathematical material and unstated conventions. We formulate faithfulness evaluation as a context-aware *align-then-label* problem: identifying phrase-level correspondences between informal and formal statements and assigning a faithfulness label to each under explicitly specified contextual interpretations and assumptions. We also introduce FaithAlign, the first theory-scale faithfulness benchmark comprising 460 items and 4,982 annotated phrase-level links, alongside FaithAlign-Gap, a companion set of 146 items with informal context and assumptions withheld. Each FaithAlign item includes the target statement pair, relevant context, formal dependencies, and, where needed, assumptions specifying one given interpretation. Across 8 models, GPT-6 Astra achieves the highest defect F1 at 47.8% and identifies all defects in only 4.5% of items with two or more defects. On FaithAlign-Gap, Astra identifies only 28.2% of undecidable items as such. These results show the difficulty of recovering context-dependent correspondences, establishing FaithAlign as a testbed for fine-grained, context-aware faithfulness evaluation.

1 Introduction

Autoformalization aims to translate mathematical writing into formal statements and definitions while preserving its meaning—a property known as *faithfulness*. This task is *theory-scale*: formalizing a mathematical definition requires tracing its dependencies on other definitions, standing assumptions, and conventions (Min et al., 2026; Feng et al., 2026). A formalization may look faithful in isolation but narrow the domain, omit an assumption, or misstate a definition in its original context, causing cascading failures in theory-scale autoformalization. Assessing faithfulness therefore requires identifying each discrepancy and tracing the dependencies behind it. When context is missing, the available information may support different interpretations. In such cases, faithfulness is *undecidable*.

Existing evaluations largely reduce faithfulness to a scalar score or binary verdict for a pair of informal and formal statements (Lu et al., 2025; Poiroux et al., 2025; Patel et al., 2026). Such outputs do not map the surrounding informal context to formal dependencies or show whether all defects have been localized. AriaScorer grounds binary judgments in retrieved library definitions but does not measure coverage of contextual correspondences (Wang et al., 2026a). FormalRx localizes problematic formal code without linking it to informal expressions and their dependencies (Wang et al., 2026b). The binary labels used by MA-Align and FormalRx cannot represent *undecidable* cases where context is underspecified (Patel et al., 2026; Wang et al., 2026b). A definite verdict thus assumes an interpretation unsupported by the supplied material.

https://arxiv.org/abs/2606.26525 https://arxiv.org/abs/2608.15432 https://arxiv.org/abs/2610.18498